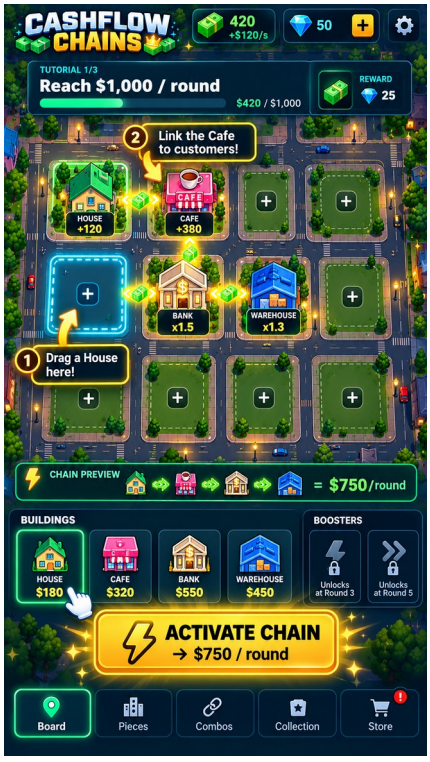
CASHFLOW CHAINS — THE COMPLETE GAME BIBLE

Final experience, retention, economy and monetization design
Edition: 23 October 2026
Product: Cashflow Chains 2.0 · Unity 6, pure 2D, mobile-first, Android then iOS
Repository: https://github.com/zaynvirkk/Cashflow-Chains
Decision status: Design direction and implementation requirements. Numeric tuning and commercial outcomes remain hypotheses until measured.
Audience: Product owner, game designers, economy designers, visual/interaction designers, engineers, LiveOps operators and Codex.

THE PRODUCT IN ONE SENTENCE
Build satisfying, ingenious chains that turn a miniature city into an economic engine; turn the rewards into a growing personal empire; then return to develop, recruit, collect, optimize, compete and discover what becomes possible next.



Provided visual direction: illustration of the proposed Cashflow Chains interface; numeric HUD values are not verified existing game-state data.

DOCUMENT MAP

Part I — The complete player experience: product vision, puzzle rules, intuitive onboarding, tactile interactions, ownership, the empire, reward loops, first and repeat purchases, retention, events, screens, analytics and final player-journey acceptance.

Part II — Implementation and economy specification: production contract; asset, currency and source/sink definitions; businesses and managers; chests and recruitment; merchants and exchange; VIP and cards; LiveOps; offers; retention; tutorial unlock graph; 41-screen matrix; tactile art rules; balance; API/backend; billing; analytics; release and test matrix.

Codex handoff principle: Part I explains what must feel true to the player. Part II specifies the authoritative systems and acceptance criteria. The two are complementary and both are normative.

How to read this document

This is a **standalone product bible**. Part I explains the game from the player's point of view, down to controls, decisions, reward presentation, progression, identity, first purchase and returning months later. Part II is the normative implementation specification: content definitions, economic ledgers, reward tables, offer and LiveOps engines, backend authority, exact screens, testing and Codex's execution contract. Where an example uses an invented figure, it is labeled **illustrative** and is not an assertion about the existing build or the reference games. The existing campaign remains reproducible and its verified puzzle results must not become pay-to-win.

Priority order: (1) an immediately enjoyable, legible puzzle; (2) a persistent account the player wants to improve; (3) rewarding decisions every session; (4) naturally relevant purchases; (5) repeated high-value experiences; (6) measurable, sustainable profit. Revenue is the commercial objective, but an empty or irritating game cannot support it. The objective is long-run player lifetime value after acquisition, platform fees, support and infrastructure—not a superficial count of store popups.

PART I — THE GAME THE PLAYER EXPERIENCES

1. Product vision: What the player actually believes they are doing

The player is not learning a list of currencies and buying virtual products. They are building their own thriving city of connected businesses. A house attracts customers; a farm supplies goods; a café serves them; a warehouse routes supply; a bank amplifies profitable flow. Pieces move with the finger, delivery arrows click into place, goods travel along real routes, and a chain erupts in understandable cashflow. The delight comes from a **cause the player can see and a consequence they can predict**. They can understand a simple connection in seconds but develop considerable mastery over routing, placement, bottlenecks and multipliers.

After the first victories, the player discovers that this is more than a one-off puzzle: the cashflow supports businesses they permanently own. Their café can become a franchise; their city can gain districts; managers can specialize in logistics, research or sales; contract routes can run while the player is away; collections can be completed; a merchant might stock the exact component needed for the next promotion; an upcoming event might reward upgrading now rather than later. What they choose matters to their long-term account. The empire becomes *their* creation—not a temporary board reset after each result.

Emotional promise, in order: "I caused that." → "I discovered a better way." → "This belongs to me." → "I have a plan." → "Something good is ready." → "I can do more." The game has to earn each transition through genuine gameplay, not simulated celebration.

Design pillars:

1. **Touch is satisfying.** Every successful placement, rotation, improvement, chain flow and collection has crisp visual, audio and tactile feedback.

2. **The economy makes sense.** Players see why a connection works, what resource is missing, and what an upgrade changes. Useful complexity is progressive, not dumped into the first session.
3. **Ownership persists.** Franchises, managers, districts, blueprints, research, earned cosmetics and achievements survive across sessions.
4. **Multiple paths, one coherent world.** The same reward item has understandable sources in free play, contracts, chests, merchants and optional purchases; it is not a different currency on each screen.
5. **Players choose their pace.** They can play short puzzles freely, save for events, optimize their empire or spend to accelerate certain goals. There is no artificial paid gate to basic puzzle interaction.
6. **Progress creates new decisions.** Promotions unlock abilities, not just larger numbers. New content generations add strategic choices without erasing the value of older purchases or achievements.
7. **Earning and buying remain honest.** Real probabilities, actual timers, exact products and predictable restoration. No fake near-misses, fabricated competitors, disguised real-money prices or counterfeit urgency.

1.1 What exists already, and the required change

The repository already provides a deterministic **eight-beat board resolver**, **32 building types**, **104 authored levels** (72 core and 32 optional expansions), **16 flow-verified combos**, **eight collection sets**, **15 one-time contracts**, local saves and a basic nine-product billing catalog. The observed code has finite puzzle rewards and narrow premium sinks. The screenshot presents a charming aspirational layout, but top-bar money per second is not proof of a currently implemented persistent idle-income system. The revamp must provide a real production subsystem if it displays live production claims.

Preserve: the board and its logic; free forecast and undo; real dependency on customers, goods and energy; correct paths, item conservation and deterministic scoring; existing levels, expansion ownership, blueprints, art rights and useful interaction code. **Replace:** the short-lived star-to-coin economy, one-time-only contract model, flat inventory, store-only monetization, limited collections, narrow rerolls and monolithic navigation. A game with a hundred products but no persistent personal goals would fail this brief.

2. The basic game, explained like a player would explain it

“Put down buildings, connect the right businesses, and press Activate to see whether your city makes money.”

Every board has a goal and a small set of buildings or building choices. The city is a grid separated by roads. Buildings have directional output and input ports. A home creates people, a supplier produces goods, and some businesses need both customers and stock to generate money. A bank can increase the eligible output of neighboring commercial activity. Energy and routing introduce new systems after the basics are clear.

The player selects a building, drags it to an eligible plot, watches it snap into the grid, and rotates it to face another building. A ghost outline shows legal versus illegal placement. Connected arrows light up when the routing is valid. A **free, exact forecast** shows what would happen if the round were activated. The player can undo, experiment and improve before committing. When they activate, the simulation advances through **eight deterministic beats**, transporting resources and resolving actual sales. The game visualizes those real events as moving people/goods/energy tokens, cash pulses and combo celebrations. The result explains what worked, what was blocked and which resources were wasted.

2.1 The rules the player must intuitively learn

Player intuition	Underlying rule	Immediate visual feedback
“Homes send customers.”	A residential source emits people along its active outgoing ports.	Small human icons emerge and follow the real edge.
“A café needs customers and stock.”	A converter consumes valid people plus goods; no invisible supply.	Missing input icons dim; eligible orders animate when both arrive.
“Arrows matter.”	Ports and rotation determine whether neighboring buildings actually connect.	Ports rotate with the building; valid links glow; invalid links do not fake delivery.
“Storage is useful.”	A router or warehouse can redistribute eligible goods under capacity constraints.	Multiple destinations receive bounded goods tokens.
“Bank makes profitable chains better.”	An amplifier applies the documented, capped modifier to eligible events.	A source-bound glow reveals which revenue it actually amplified.
“Longer isn’t always better.”	Limited beats, edge capacity, resource budgets and bottlenecks limit efficiency.	Slow, spilled or unserved resources are visibly identified.
“Better planning makes more money.”	Improved layout raises legitimate deterministic profit and/or fulfills harder	Forecast delta and result breakdown show the change.

Player intuition	Underlying rule	Immediate visual feedback
	objectives.	

The board must not lie. If a café needs goods, the first tutorial must visibly supply or pre-stock the goods. If a building is out of budget or its route is invalid, the UI must state the specific reason. If money rises on-screen, the source must be a genuine sale or reward, not a visual particle granting unearned currency.

2.2 Three separate kinds of money

A major source of potential confusion is displaying "$750/round," "$120/s," "420 cash" and premium Gems as though they are the same resource. The final UX must distinguish:

- **Round Cashflow / Score:** the score from the eight-beat simulation, useful for stars, mastery and comparison. It is not a persistent wallet, nor does every replay mint its full value again.
- **City Cash:** the persistent earned soft currency used on owned businesses, construction, research and certain merchant trades. The reward engine defines capped grants from eligible activities.
- **Spark Gems:** premium virtual currency, earned sparingly through genuine milestones and optionally purchased, used for explicit convenience, draws, refreshes and some packs. No real-money redemption is possible.

Use icon, color, unit and textual context to distinguish all three. A label such as **+$120/sec** appears only when there really is a measured, capped production rate over a known collection window; show the underlying source and storage limit. Never use fictional performance as a promise of real-world income.

2.3 The board has two responsibilities—not one muddled mode

Certified puzzles preserve fixed rules and replayable skill challenges. All players see an identical eligible challenge. Owned/purchased power cannot modify that challenge's scoring or solutions. Players unlock stars, cosmetics, mastery and bounded empire rewards through genuine performance.

Empire operations reuse the same chain mechanic but allow the player's owned businesses, managers, equipment and research to affect output. These puzzles can reward persistent resources, franchise experience and events within server-enforced caps. This is where upgrading an owned business is valuable. A successful empire can grow more powerful without destroying the logical integrity of fixed campaign puzzles.

Other modes—dispatch, daily skill, events, cooperative projects and leagues—must declare whether owned power applies. Never mix a pay-influenced empire leaderboard into a supposedly equal skill competition.

3. The first impression: Every gesture must demonstrate value

The opening should avoid a long title cinematic, dense business language, account registration or nine visible monetization systems. A player should interact with the board almost immediately, understand one action, and see a real improvement caused by that action.

3.1 First-session sequence is trigger-based, not fictional day numbers

Entry. Show the miniature city, the target and only the essential controls. A short animated delivery pulse can demonstrate the system, but no account currency is granted until a legitimate event is completed. The first instruction points to a house/café connection. The player moves the relevant piece themselves.

First drag. Lift the selected building 5–8 pixels, soften its origin, show a floor-aligned ghost, highlight valid tile outlines and let the card magnetize to the target plot. An invalid spot yields a soft recoil and a readable explanation. On successful drop, apply a fast snap and a small cascade along newly connected roads. The city visibly responds before any dialogue box opens.

First forecast. Show "Before: 0 / After: [actual deterministic projection]" beside the objective. The connection light and input icons explain the improvement. Don't immediately hide the result under a reward offer.

First activation. Make the eight beats short, legible and skippable. Tokens travel the routes that really exist. Sales animate at the café; money moves to the result counter. Show one clear line linking the player's action to a payoff. Award a small first-clear reward and an earned standard chest once the server/local authority commits it.

Second concept. Introduce rotation in a new board. The player chooses the orientation; the UI teaches why changing one arrow creates a valid connection.

Third concept. Introduce multiple inputs or routing. The player combines placement and rotation and achieves an improvement they can recognize. Retain free reset, forecast and undo.

First personal investment. Give the player a permanent starter business, and show that it has a location and a future development path. They can name their city and select an earned initial accent/theme without interrupting puzzle play. Do not hand them a supposedly top-tier manager before teaching what managers do.

First significant earned unlock. After the player has demonstrated placement, rotation, a combo and a short commission, award an appealing Rare/Epic franchise or manager with substantial further optional development. Unlock its detail screen, show stars, actual fragments owned, remaining requirements, free acquisition routes and clearly useful next capabilities. The unlock is milestone-based—not automatically on calendar Day 1.

First commercial opportunity. After the player has used a real builder or contract slot, reveal the permanent second-slot option near the system it helps. Explain both the paid and earnable alternatives. A small starter offer may be eligible after clear engagement, but it should not interrupt the reward claim, block continuation or claim that an offer is expiring unless its deadline is real.

First recurring systems. Unlock recruitment only after the player has a reason to recruit. Give an earned first ticket. Unlock the merchant after the player understands material acquisition. Unlock events after the player has a repeatable activity they can contribute to. Introduce VIP after the player has seen one of the underlying benefits. Introduce equipment and advanced currencies later.

The exact level numbers and calendar timing for these triggers belong in versioned configuration and are tested with fresh accounts. Tutorial success and content unlocks—not elapsed days alone—are the primary gates.

3.2 What the player should feel at different points

Moment	Player's question	Required answer from the game
First 15 seconds	"What do I do?"	One glowing place-to-connect action, not five icons.
First minute	"Did I cause that?"	Visible before/after cashflow explained by a real route.
First few minutes	"Can I get better?"	Rotation, a surprising combo, improved forecast and a harder board.
First 10–20 minutes	"Does this matter tomorrow?"	Permanent business ownership, tangible upgrades and renewable contract progress.
After first major unlock	"How do I develop this?"	Named free fragment sources, progression milestones and optional acceleration.
First next-day visit	"What has changed?"	Real completed timer, bounded production, refreshed objectives or earned claim.
Returning week	"What should I focus on?"	Several viable paths: franchise, manager, district, event or collection.
Returning month	"Why am I still here?"	Identity, specializations, mastery, new content and a personal history of achievements.

4. Make interaction itself delightful, not only the rewards

Cashflow's miniature city and kinetic chains are the primary aesthetic product. Visual polish cannot be postponed until monetization is done: movement quality affects whether anyone stays long enough to encounter the economy.

4.1 A grammar of touch feedback

Tap. On pointer-down, the control compresses subtly and its highlight strengthens; on release, it rebounds and emits a restrained click. Feedback must be immediate even if the server will later confirm the reward. If a network operation follows, show a stateful spinner or pending label without pretending it has succeeded.

Press-and-hold. Show a progress ring, subtle elevation, haptic tick and context preview. Long-press should reveal optional detail, not be required for essential understanding. Holding a building reveals its input/output ports, production limits and current missing resources.

Drag. The selected building rises, casts a flat 2D contact shadow, and follows the fingertip with a tiny intentional lag. Nearby valid lots anticipate the object; a target cell enlarges slightly. A legal landing produces magnetic snap, spring compression and an energy pulse. An illegal drop reverts cleanly with a soft, non-punitive cue. No accidental charge for dragging or Undo.

Swipe. Use horizontal swipes to navigate adjacent collections or shop tabs; vertical swipes to browse clearly scrollable lists, not rotate an important resource. Card decks use consistent momentum and snap. Never trap players in ambiguous gesture-only interactions; provide visible buttons as alternatives.

Rotate. The asset rotates exactly 90° in a short easing motion. Ports move with the art, affected routes re-evaluate and the forecast delta animates only after the actual resolver has recalculated. A valid newly connected edge sings a rising note; a disconnected edge fades silently.

Activate. The signature moment. The button compresses, emits a rich pulse through valid roads, then resolves the actual eight beats. Goods, people and energy have distinct icon/shape/color treatments. Profit flows toward emitting businesses and then the result HUD. Multiple legitimate transactions create rising sound layers; combo events become larger but never obscure the source route. A Skip button always yields identical stored results.

Collect. A reward card opens, the actual item is revealed, and a small icon flies to the proper wallet or inventory destination. The counter changes exactly once from an authoritative grant. If the item finishes a promotion or set, show a secondary milestone as a consequence, not a fabricated reward.

Upgrade. Resources leave the inventory visibly, the chosen building improves with a concise evolution animation, the newly unlocked ability is explained in concrete terms, and the next requirement appears. Players can compare before/after stats rather than trusting a vague “Power +999.”

Open chest. Lock movement during a brief presentation, not during reward determination. Give a textured click, short light reveal, rarity-appropriate sound, and item cards. Allow Skip, Open 10, Claim All and reduced-motion presentation. Outcomes are already determined and committed before animation; do not fake near-misses.

Gacha reveal. Use different presentations for ordinary fragments, new complete managers, duplicate conversion and guarantee rewards. The revealed result must match published odds, pity logic and immutable server settlement. Show the transaction history and remaining pity progress after the claim.

Merchant refresh. One swipe or button swaps the shelf cards in a coherent flip/sweep animation. The game discloses any cost before execution. An unsuccessful networking request retains the old stock and currency. Paid rerolls sample the published pool; do not falsely promise an upgraded rarity.

Claim pass/event. Progress bars fill from verified qualifying actions. The player’s legitimate milestone can unlock one or several track rewards; visually separate the free and paid claims. Claim All does not require multiple confirmation screens.

4.2 Practical interaction targets

Interaction	Initial motion target	Hard UX principle
Button down/up	70–130 ms response + spring	Visual response starts immediately; never blocks input invisibly.
Tile snap	120–220 ms	Must align with actual legal coordinates.
Rotation	130–210 ms	Hitboxes and ports update consistently.
Counter increment	200–450 ms	Count matches transaction; never exceeds authoritative value.
Standard chest	About 1 second	Skip and accessibility alternatives always available.
Major promotion	About 1.5–2.5 seconds	Rare enough to retain significance.
First chain resolution	Readable under a few seconds where feasible	Player can inspect cause and effect; later speed controls available.
Navigation	120–240 ms	No blocking decorative transitions on repeated visits.

These are **design starting targets**, not strict physical laws. Profiling on real phones, reduced-motion testing and touch accessibility take priority over animated flourish. Aim for a stable 60 FPS on mainstream hardware and a graceful 30 FPS on lower-end devices. Audio layers must be ducked/mixed; motion, sound, haptics and flash intensity must each have independent controls.

4.3 Visual identity

Pure **orthographic 2D** city board: no isometric buildings, no 3D meshes, no perspective roof illusions. Keep the screenshot’s bright miniature-city energy while making the playable geometry technically honest. Palette: midnight teal/indigo streets,

vivid mint flow, warm amber cash, high-contrast white type, strong distinctive building silhouettes, real lighting accents and restrained sparkles. Make the building art read at phone scale. The top HUD must reserve space for the objective, the chosen premium currency, account progression and a small number of current system statuses; no crowded mobile spreadsheet. UI typography is native and localizable, not baked into sprites.

The actual city has life even when the player is not interacting: limited cars move along correctly aligned lanes; tiny pedestrians respond to active businesses; windows illuminate, fountain/power equipment animates, delivery routes pulse, and each business family has distinctive ambient sounds. These effects are bounded, never simulate extra gameplay rewards, and obey reduced-motion settings.

5. Make the city belong to the player

A return-worthy game needs a persistent object of care. A set of completed levels is progress, but it is not enough of a personal identity. Cashflow Chains must offer several forms of ownership, all visible and consequential.

The city: The player names a city, selects a neighborhood identity, chooses a visual theme, and watches an empire map physically fill with businesses they developed. Districts have recognizable roles and visual evolution: first street, neighborhood retail, logistics, industrial, financial and advanced districts. The city map should be a **trophy and planning tool**, not a full NPC simulator. A player can share a picture of their unusual cashflow layout or city evolution without exposing personal data.

The franchises: A discovered piece recipe (e.g. Café) is not the same thing as an owned, upgrading franchise. The recipe defines legal puzzle behavior; an owned franchise has its own level, star rank, manager assignment, equipment, production capacity and skin. Different franchises serve different objectives, so a player can decide to specialize in food, logistics, finance or a balanced economy. Each has an actual job in the persistent world.

The managers: Manager characters make the city feel inhabited without expensive AI simulation. Give each manager an identifiable role, expression, readable ability and simple backstory conveyed in short inspectable cards. An Operator increases throughput for qualifying business families; a Developer aids construction; a Logistician optimizes dispatch; a Financier changes profit or conversion behavior. Manager ownership should yield both playable decisions and visual identity. A hero collection with no functional distinction is only artificial scarcity.

Blueprints: A solved layout can be named, saved, improved and replayed in eligible modes. The player sees their own inventions as artifacts. Blueprints record validated piece identities, ruleset versions and intended mode, not purchased scoring exploits. Saved layouts can unlock legitimate quality-of-life conveniences; core editing and Undo remain free.

Collections: Albums, building families, manager sets and district trophies provide visible partial progress. Completing a set awards a known bonus or appearance. Duplicate fragments convert at clearly stated rates; common pieces may be exchangeable or giftable where a genuine social feature exists. A collection should never be portrayed as complete until it really is.

Achievements and history: Let players inspect their first successful chain, highest legitimate per-round cashflow, most efficient route, first promoted franchise, completed events and city anniversaries. Distinguish earned accomplishments from premium ownership. Personal history supports attachment without punishing breaks in play.

5.1 The player must make decisions worth caring about

Instead of a linear “Upgrade everything” button, resource allocations should create real opportunity costs. A player with enough material for one project decides between another contract slot, a new warehouse level or the next café star. Those choices change what the player can do next. Clear opportunity costs make progress feel earned, and help the player form their own strategy.

A specialist path must be viable: one player builds a high-throughput logistics empire, another uses excellent retail chains and collections, another optimizes dispatches, another pursues mastery boards. No single purchase should make every previous decision irrelevant. A visually distinct city plus a mechanically distinct development path is what makes the account feel personal.

6. The persistent empire: The actual reasons to come back

The empire is the state layer that turns short puzzle victories into a multi-month journey. It has assets, production, construction queues, workers, research, operating contracts, storage, collections and its own world clock. It does not run an unbounded idle formula that generates infinite money while the user is gone.

6.1 Business lifecycle

Discover → Acquire → Build → Operate → Upgrade → Promote → Specialize → Master. Each step has a separate reason to exist:

Lifecycle step	What changes in play	Persistent demand or opportunity
Discover	The player learns the recipe and how it combines with others.	Complete meaningful boards; collect mastery records.
Acquire	Own a permanent franchise card/asset.	Initial fragments, achievement, event or earned choice chest.
Build	Place the franchise in the permanent empire map.	Cash, construction materials, builder time.
Operate	Earn capped production or complete eligible operations.	Managers, operating slots, storage and contracts.
Upgrade level	Increase a clear baseline capability.	Cash, resources and recurring construction queue.
Promote stars	Reach an ability breakpoint.	Hero-style specific or universal fragments.
Specialize	Select a defined branch/operating talent.	Research and specialist resources.
Equip	Install machinery that solves a different weakness.	Equipment parts, cores and mastery resources.
Master	Earn an advanced identity and unique but bounded benefit.	Rare milestones, collections, later events.

Illustrative Café example: At acquisition it serves as a basic retail franchise. At a subsequent star rank it may raise the number of customer orders that can be served in designated empire operations; at another rank it can tolerate supply interruptions or generate a small bonus on qualifying Morning Rush combos. Equipment may increase production efficiency, while the manager affects dispatch yield. On certified campaign puzzles, none of these account-wide bonuses override the reference score.

6.2 Managers and jobs are deliberately different

Managers have rarity, stars, levels, XP, skills, equipment and assignable roles. A star or level unlock must be visible in their card, with a concise “before / after” description. An acquired rare manager should not require spending to remain useful. A higher rarity can provide an interesting skill, not merely a bigger arbitrary number. Universal fragments are valuable because they resolve the player’s own selection problem; scarce manager-specific fragments give long-term direction.

One manager can serve multiple businesses only if the employment rules permit it; an assignment capacity introduces a genuine planning decision. Preset slots and convenience tools simplify management, while base reassignment stays accessible without charging players to correct innocent mistakes. Avoid duplicating twenty progression bars for every manager unless each makes observable gameplay difference.

6.3 Construction and time

Construction is a meaningful empire operation, not a delay before the next puzzle. One builder can upgrade one building; an additional builder can work on a second valid project. A player can wait, earn speedups or optionally accelerate. The same rules apply to research and dispatch with distinct timer categories.

A **permanent second builder** is the cleanest candidate for an early utility purchase because its value is legible after the player experiences a busy queue. It is not automatically the best-priced or highest-converting SKU; that must be tested. There must be an earnable alternate route to basic functional progression and no hidden timer inflation after ownership.

6.4 Production, storage and offline returns

Persistent production must have a real rate, a source, a finite storage capacity and collection rules. Simple version: every owned operating asset contributes a capped resource production amount per hour, conditional on researched modifiers and valid assignments. Offline earnings are calculated using a trusted elapsed window and a storage cap; no arbitrary device-clock forward trick can generate valuable premium resources. The home screen shows “ready to collect” or a precise timer, not an invented reward notification.

The first day should offer valuable short-term activities even while construction is underway. The minute/hour return loop can be driven by real dispatch completion and production collection, but neither is allowed to hold puzzle gameplay hostage.

6.5 Contracts: Earnable recurring resource pipelines

The current code has fifteen one-time commissions. Rebuild them as a renewable dispatch framework using the player’s validated saved chain layouts. Each contract is defined by requirements, time, eligible business family, risk or modifier, payout pool and completion ID. Players select a contract and a qualified chain; the domain verifies the layout, then a trusted clock schedules completion. On claim, the result grants materials, fragments, cash, experience or a chest, and advances applicable mission/event tracks exactly once.

Offer multiple valid strategies: short dispatches for active players, longer contracts for less frequent players, specialized contracts yielding rarer fragments, and limited contract slots for capacity management. Contract completion should feel like the city continues operating, not like a mobile timer the player bought the right to stop looking at.

7. The economy is a graph of needs, rewards and alternatives

One important rule: never define a purchase solely as “what it gives.” First define the item, the upgrade it serves, its free earning routes, how frequently it is needed, and the alternative use of the player’s time. Then define how that item can appear in a merchant, chest, event or bundle. That is how commerce becomes a natural part of the game.

A material such as Café Fragments has several acquisition edges and one or more useful consumption edges:

Eligible puzzle / dispatch / event / daily objective → fragment or earned chest → inventory → Café star promotion → stronger operating choices → harder contracts and new events.

Other acquisition edges may enter the same inventory: earned trade token → exchange, rotating merchant stock → Cash or Gems, recruitment/selection chest → fragment, or optional paid pack → fragment. The player sees these via an accurate **Find Sources** sheet directly on the upgrade screen.

7.1 Currency hierarchy

Keep the visible default HUD simple: persistent Cash, Gems and at most one relevant event currency. Place the rest next to the activities where they matter. Use descriptive category labels for materials, fragments, XP, gear, speedups, tickets and event currencies. Each should have stable IDs, a reason to exist and an accountable ledger. Basic resources remain earnable repeatedly; specialized endgame materials unlock when the player can actually use them.

Sources include real chain achievements, capped open-city production, contracts, milestones, free passes, event tasks, trading, chest loot and occasional free Gems. **Sinks** include business construction, franchise stars, manager promotion, research, gear, production upgrades, merchant refreshes, recruitment and event participation. The system must prohibit profitable closed-loop currency arbitrage, hidden negative balances, replay farming and inflation that trivializes progression.

7.2 The powerful connection: One upgrade creates another useful choice

A second builder lets the player work on two assets; this consumes materials faster and makes contracts or production more useful. Stronger managers improve certain operations, which produce resources for new upgrades. Specialized equipment improves a different attribute than star ranks. Research improves processes and unlocks new routes. Events temporarily reward a subset of normal actions, so timing the consumption of saved materials can be an actual strategy.

This coupling should be **complementary, not a trap**. An asset must remain usable before all six sub-upgrade trees are maxed. Players should be able to see the benefit of the next step and the alternatives, instead of encountering a chain of unexplained, ever-new blockers.

8. Design the reward process, not only the reward amount

The game's collection and chance systems need both **uncertain results** and **predictable progress**. They should create moments worth opening the game for, while remaining honest about what the player may obtain.

8.1 The reward formats

Format	Player action	What is emotionally/strategically different	Required safeguard
Direct reward	Claim earned cash/materials.	Predictable progress; low friction.	Exact amount and grant receipt.
Earned chest	Open a container earned in gameplay.	Surprise and discovery.	Visible category and real loot table.
Premium chest	Acquire a paid chance-based container.	Additional optional access to reward pool.	Odds disclosed before purchase; age/region checks.
Choice chest	Select one of known rewards.	Agency and shortage resolution.	Item only consumed on confirmation.
Themed chest	Draw from franchise/manager/gear pool.	Relevant pursuit.	Complete pool and effective odds.
Chest merge	Exchange multiple lesser containers for a stronger one.	Real "open now or save?" decision.	Explicit conversion and expected-value consequences.
Recruitment	Draw managers/fragments and build pity.	Collection and recurring character goals.	True odds, guarantees and duplicate rules.
Milestone chest	Reach a known activity or draw count.	Certainty following effort.	Actual threshold and persistent progress.
Fragment album	Assemble a picture/set from earned pieces.	Completion and personal investment.	Duplicate conversion and real progress.
Merchant shelf	Inspect a rotating choice of items.	Chance before selecting a purchase.	Stock and refresh cost transparency.

These are not ten independent codebases. One item catalog and one verifiable reward engine underlie them. Reveal animation is a presentation of already determined results. The code should support Open 1, Open 10, Skip and Claim All with consistent correctness and speed.

8.2 A gacha system players can reason about

Recruitment may use earned tickets or Gem draws. A featured banner displays real chances, duplicate conversions, guaranteed draw milestones and a pity counter. New managers have visible abilities worth using before max promotion. An ordinary player should obtain some recruitment and promotion progress from free play; paid draws accelerate progress and offer more optional acquisition chances.

An illustrative initial banner in the technical specification uses a 2% complete Legendary outcome, a guarantee by 80 eligible draws and known 10/30/60 draw milestones. **Those are proposed simulator values**, not evidence of optimal commercial odds. Before a live offering, calculate effective probabilities for any pity or featured mechanics, show them to the user, and test whether the free route creates real progression.

No near-miss wheel that falsely depicts almost landing on a rare result, no dynamic hidden luck boosts targeted at a frustrated player, and no fake "99% lucky" language. The game's excitement should come from genuine acquisition and meaningful ability unlocks.

8.3 Rotating merchants turn acquisition into a game

A shop can be a gameplay surface rather than just a product list. The Daily Merchant has six visible slots sampled from an eligible inventory pool; a real refresh period; one earned/free manual refresh; further explicit voucher or Gem costs with caps; stock limits; and a visible way to compare expected usefulness to the player's current goals. VIP and event merchants use the same framework with different currency and stock rules. If a merchant reroll costs premium currency and changes random stock, display sampling odds before the spend.

Refreshes must not deceptively claim to improve rarity when they merely resample. A shop can occasionally have a published guarantee after a set of refreshes, but that condition must be real and carried in persistent state. Shop prices show actual

items and quantities, not misleading fake list-price discounts. A failed purchase or connectivity issue must never erase existing stock or funds.

8.4 The “almost finished” feeling should be earned, not rigged

A visible next-star requirement, an incomplete album and a manager needing a few genuine fragments create clear voluntary goals. There should be plenty of normal states where a player is *not* close to completion; do not secretly manipulate drops so someone is perpetually one item short. A player may choose to earn the final fragments through an event, select them from a choice chest, wait for the merchant or buy an appropriate bundle. The best experience gives several honest paths to an outcome they genuinely want.

9. What creates the first purchase

The first purchase is not won by showing a $0.99 banner in minute one. It is won when the player recognizes a **useful, persistent or immediately relevant benefit**, trusts that the product is real and knows the game will continue to be enjoyable afterward.

9.1 First-purchase candidates ranked by clarity, not unproven conversion

Permanent second builder. The player has already started an upgrade and wants to develop another business. A second queue makes its benefit obvious in the exact moment of need. Show remaining wait, alternate earned route and exact permanent effect.

Small starter package. After multiple successful puzzles and a first tangible owned franchise, an optional entry pack can combine useful materials with a cosmetic identifier and perhaps the low-price unlock of a relevant starter manager. The manager itself must still have a clear development path; the pack does not need to supply everything.

Monthly convenience trial/card. Once a player has used repeatable contracts and production, daily automated claims or modest additional capacity may be compelling. Do not sell automation before the player has experienced the corresponding manual operation.

Currency package for an explicit objective. If the player voluntarily pursues a featured recruitment milestone or rotates a merchant, a small Gem package may be relevant. Do not create hidden confusion about exchange rates or default to paid currency when earned vouchers remain.

9.2 First-purchase funnel: Measurable events

Feature used → repeated legitimate need → product detail voluntarily opened → value compared with free routes → platform price displayed → confirmation → verified fulfillment → visible benefit → subsequent play.

Measure each step and the retention that follows. If a second builder is purchased but the player churns immediately, short-term conversion alone doesn’t establish a successful design. Track refunds, regret indicators, future sessions, future spending and whether they continue to use the unlocked capacity.

9.3 First purchase must never feel required to “really start”

The first purchase may accelerate progress but cannot be necessary to solve a free certified puzzle, undo a mistake, understand an explanation or participate in normal early gameplay. The system should provide a correct free route and clear value. Player confidence is more valuable than a single angry transaction.

10. Why players pay again and again

After first purchase, different needs emerge at different moments. A permanent unlock does not replace consumable materials; a monthly card does not replace a season pass; a good manager does not eliminate equipment and specialization. The key is that **the products are complementary because the underlying gameplay activities are complementary**.

10.1 Revenue portfolio by relationship to progress

Product family	What it changes	Repeatability	Context where it makes sense
First-purchase pack	Entry utility + meaningful introductory resources	Once	After the player understands the included systems.

Product family	What it changes	Repeatability	Context where it makes sense
Permanent builder / utility	Parallel operations and convenience	Once per defined unlock	A legitimately occupied queue.
Gem packs	Flexible optional purchasing power	Repeatable	Choosing among Gem-based acquisition routes.
Construction/dispatch speedups	Time to completion	Repeatable	Current queued work or timed event.
Business fragments	Progress on an owned franchise	Repeatable	Specific star upgrade goal.
Manager/recruit packs	Additional character development	Repeatable	Recruitment feature and desired roster.
Skill/equipment materials	Improve already-used characters or machinery	Repeatable	Equipment or skill breakpoint.
VIP-specific pack	Tier-relevant rewards plus progression	Staged	Newly eligible VIP levels.
Weekly card	Short recurring supply stream	Renewable	Player regularly operates resource systems.
Monthly card	Predictable daily utility/rewards	Renewable	Returning player with routine activity.
Season pass	More rewards for completed real tasks	By season	Active, genuinely completable season.
Growth fund	Milestone-linked guaranteed account rewards	Stage-specific	Owned account reaching relevant thresholds.
Event packs	Extra material or tickets for a chosen event	By event	Player is actually participating.
Choice bundle	Resolve a chosen bottleneck	Repeatable	Several material paths are available.
Sequential purchase ladder	A disclosed series of purchases with bonuses	Event-limited	Clear total cost and no hidden future requirement.
Cosmetics / identity	Player's owned world looks different	Collection-based	Display, customization and social recognition.

10.2 A single purchase can feed several honest systems

Suppose a player buys an eligible manager bundle. The initial transaction grants stated fragments and XP. If the purchase has advertised VIP XP, that advances VIP. If a published top-up milestone is running, the eligible transaction advances its counter. If the game has a real alliance and the pack explicitly grants group gifts, genuine teammates may receive those gifts. If the player then uses the fragments during a manager-development event, **spending the material** may separately generate earned event points. Each step has a unique grant/score ID and cannot be double-counted by retries.

The perceived value is multi-layered because the game rules are interconnected, not because a popup invents six free gifts after payment.

10.3 Rotations and account-specific catalogs

An eligible offer can change with account progression, unlocked features, content generation, event participation, already-owned permanent goods, region/platform price, previous purchases and deliberate specialization. The offer is not necessarily personalized by a secret willingness-to-pay algorithm. Start with deterministic entitlement and relevance rules that players can understand, then test catalog composition and placement ethically.

Offer types should rotate through a published or accurately represented LiveOps schedule. A material relevant to Research Week may be featured when Research Week is active. A new manager banner and matching fragments may become available when its cohort's generation opens. A player who owns a second builder should no longer see that SKU as though it were an upgrade they can buy again.

10.4 High-spending users need depth, not infinite coercion

Some players choose to spend heavily on collection or competitive development. Give them meaningful advanced projects, specialization, complex events, optional identity, satisfying powerful assets and clear premium value. Do not assume unlimited ability to buy a leaderboard position is harmless; cap competitive advantages where appropriate and clearly distinguish spend-influenced leagues from pure-skill rankings. Extremely aggressive economics can erode the active free community that makes competition meaningful.

10.5 Purchase timing without intrusive interruptions

Persistent contextual availability: Find Sources links, merchant menus, contract/upgrade details and the Store tab may be accessible whenever eligible. **Rare proactive surfaces:** first relevant feature introduction, genuinely new generation, a selected active event or a return after a significant account change. Never repeatedly block the board, results or free reward claims with aggressive paywall modals. A daily bonus should be discoverable without making the player dismiss it before playing.

11. Retention: Give the player meaningful reasons at every timescale

Retention is not one daily-reward button. A long-lived game has several nested return loops with different motivations, and each must be backed by an actual improvement, decision or opportunity.

Timescale	Trigger	What the player looks forward to	Related economy
Seconds	Placement, rotation, reveal	Sensory reaction and correct causal feedback	No transaction required.
Under a minute	Link correction, chain activation	Seeing a better outcome from an action they chose	Skill, chain score and round rewards.
A few minutes	New layout, combo, commission	Applying learned rules and getting something new	Cash, XP, materials, chests.
10–30 minutes	Puzzle series, dispatch/upgrade decisions	Completing self-chosen goals and improving the account	Franchise, managers, contracts, collections.
Hours	Building/dispatch ready	Real project completion and next allocation decision	Construction, research, production.
Next day	New daily mission or real shop reset	A fresh configuration or reward opportunity	Merchant, VIP gift, daily tasks.
Several days	Weekly event or saved-material moment	Strategic progress toward a medium-term target	Event tokens, milestone rewards, guild.
Weeks	Season objective and larger unlock	Visible ownership milestones and new abilities	Season pass, new district, franchise mastery.
Months	Content generation, prestige milestone	A world that changes while respecting previous work	New managers, specialist systems, city history.

11.1 Three reliable reasons to return

Completion: “My project is ready.” This must be true. Show the actual building, dispatch or capped production that completed, and let the player take the next action in one or two touches.

Curiosity: “Something different is available.” A genuine rotating merchant, an earned chest, an event challenge or a newly unlocked business can provide discovery without requiring payment. Avoid meaningless daily random noise that overwhelms player planning.

Agency: “I have a plan I want to finish.” The most durable loop arises when the player chose an upgrade target, manager, blueprint or collection and can measure progress toward it. The game should surface saved personal goals without taking control of the player’s decisions.

11.2 Daily tasks should recognize normal play

Daily objectives are not a checklist of arbitrary taps. Qualifying tasks might include completing a chain, improving its forecast, dispatching a contract, collecting production, opening an earned chest, using a merchant trade or developing a business. A

player who normally plays puzzles should make progress toward daily objectives through puzzles. A player focusing on franchise upgrades should still have relevant qualifying options. Give modest guaranteed milestone rewards and avoid a last-minute penalty for skipping a day.

11.3 Returning after an absence

Do not reset meaningful progress, erase earned streak prizes already claimed, drop owned managers or shame the player for missing a schedule. Provide a compact return report: what completed, what was capped, which saved goals remain, what new content is available and one sensible activity to resume with. Any catch-up rewards should be explicit and consistent, not a fake countdown attached to a purchase.

11.4 Notifications and respect

Notifications are opt-in and grounded in real events: a selected construction has completed, a named dispatch is ready, an opted-in event is approaching its actual end, or a requested friend/group contribution has arrived. No fabricated scarcity, fake competitive losses, “Your city will collapse!” messaging or repeated push alerts in quiet hours. Users can disable categories separately. Engagement measured through genuine satisfaction is more durable than forced re-entry.

12. Events and LiveOps: Revalue normal actions without inventing another game

An event should temporarily make some **existing** gameplay choices more rewarding. A construction event rewards qualifying upgrades; a Franchise Festival recognizes star promotions and eligible combo chains; a Delivery Derby uses real contracts; a recruitment event adds known milestones for genuinely completed draws. Only the event rules and reward catalogs change: the underlying buildings, resources and player account remain the same.

12.1 Recommended operating calendar

Event family	Valid player action	What is earned	Store/reward relationship
Chain Masters	Earn verified target cashflow and combos	Skill medals, free progress, chests	Skill boards must not gain paid-power advantages.
Build Rush	Finish real construction or use eligible materials	Development points, plans, speedups	Construction bundles may be relevant.
Franchise Festival	Promote owned business, complete its contracts	Franchise tokens, specific fragments	Choice fragments, featured business rewards.
Executive Recruitment	Recruit/promote valid managers	Milestone chests, targeted fragments	Banner tickets and disclosed guarantees.
Merchant Fair	Trade in shops and complete merchant objectives	Merchant vouchers, event tokens	Limited stock and published reroll rules.
Delivery Derby	Dispatch and claim qualifying contracts	Gear/materials, rank points	Dispatch speed, capacity, optional card.
Research Week	Finish tech projects	Research resources, progress	Research materials and speedups.
Collection Season	Finish sets, exchange duplicates	Set prizes, cosmetics, album rewards	Earned and optional fragment sources.
Empire League	Reach progress goals in paid-power-enabled mode	League rewards and prestige	Must be labeled progression-influenced.
Cooperative Project	Contribute verified resources/activity	Group milestones, shared cosmetics	Only actual people, accounts and donations.

A 28-day season can organize four weekly themes and one persistent free/premium progression track. Do **not** force a fixed event calendar into every account regardless of progression; the event engine checks content generation, feature unlocks and player eligibility. Use actual reset times and clear start/end displays.

12.2 Event stacking: The same real action can advance more than one goal

Suppose a player completes a Café star promotion during Franchise Festival while they have a season mission to upgrade businesses and a legitimate collection milestone. The permanent Café improvement, the festival points, the season XP and

the collection record can all advance. The UI summarizes the results once, rather than popping four different celebratory windows. If the player later replays an old screen or receives a retry response, it must not grant any component twice.

Important separation: Buying shards may qualify for an advertised purchase milestone; **using shards** may qualify for a development event. These are different verified actions and are not an excuse to mint points multiple times from the same transaction. When overlapping rewards get out of proportion, tune the formulas and event schedules, not the visual deception.

12.3 Content generations without destroying owned assets

Later cohorts can unlock managers, franchise branches, rare construction materials, equipment tiers, advanced research and fresh event categories. The old city still operates. Specialization means older managers and businesses retain useful niches rather than becoming garbage overnight. Content-generation timing is based on the server/cohort clock while personal onboarding is based on actual player progress; these must never be conflated.

12.4 Social without rebuilding a full MMO

The first architecture may use asynchronous cooperative groups, real group missions, giftable eligible fragments, shared event totals and separate ranked modes. The client must never invent fake players or imply that NPC contributions are real teammates. Keep privacy defaults conservative and user-generated content moderated if introduced. Real-time wars and massive world simulation are optional later investments, not required to connect the core economy.

13. A complete example of one player's evolving motivations

This is an **illustrative narrative conditional on actual achievements**, not a claim that a specific event or hero appears on a fixed calendar day.

First session: The player moves a house next to a café, sees customers flow, rotates a supplier, and witnesses a profitable chain. They claim a first-clear reward. A tiny city map shows their new business district. They make progress in a first collection without needing to understand collections in detail.

After learning the basics: The player chooses whether to save Cash for the next business construction or improve an operating franchise. They acquire an appealing business or manager through an earned milestone, see the star promotion path and get a genuine free source of additional fragments. A contract dispatch offers a reason to check back, but the player can keep playing puzzle boards immediately.

When capacity becomes relevant: They have enough materials for a second useful upgrade while the existing builder is busy. The game shows the timer, the free wait/earned speedup route and the optional permanent second-builder offer in the same context. If they decline, the current project and puzzle gameplay proceed normally.

When the merchant unlocks: They see six randomly selected but relevant goods, one free refresh and transparent voucher/Gem alternatives for further refresh. A useful fragment may be present, or not. They make a real comparison between a merchant purchase, a free dispatch and saving resources for the next event.

During the first appropriate recruitment event: An earned ticket opens a manager draw, the player sees the resulting character/fragments, the next deterministic milestone and the disclosed guaranteed threshold. They discover that an advanced manager has a role, skills, equipment and star ranks—not just a premium label. There is a reason to develop the character without needing to pay.

Over a recurring event: They save materials and plan an upgrade during the correct scoring window. The actual upgrade also advances their season track. A player who chooses to spend might purchase missing materials to reach an additional published milestone; the free player might optimize timing and earn worthwhile rewards anyway.

Months later: The city has their own look, specialized franchises, favorite managers, saved blueprints, a record of past cashflow improvements, partially completed collections and future research. New content becomes available, but the old account remains useful. There are fresh puzzles to master, new item pools to learn and several personal goals to choose between.

This is the product's retention argument: a combination of tactile joy, intelligible mastery, personal investment, useful routine activity and evolving optional goals.

14. Screen-by-screen information architecture

Use the current screenshot as **visual inspiration**, not as a command to cram every system onto one phone screen. The board is the dominant interaction surface. A bottom rail can expose Board, Empire, Managers, Events and Shop; the remaining systems live inside contextual subviews. A material icon opens a source sheet; a manager portrait opens its development page; an event badge opens the relevant actual event—not an unrelated store.

14.1 Major journeys

From first install: Boot → optional consent/essential settings → guided board → immediate feedback → verified result → first permanent reward → next board or city hub.

From the city: City → business detail → upgrade requirements → earned source/merchant/optional offer → confirm chosen route → upgrade/reward receipt → city visibly changes.

From an earned chest: Inbox/inventory → chest detail and full pool → open → server-committed reveal → inventory update → optional upgrade-ready shortcut. No paid offer needs to interrupt the reveal.

From recruitment: Managers → banner with complete odds, guarantee and progress → ticket or Gem choice → confirmation if currency is consumed → reveal → manager card / duplicate conversion → next goal.

From an event: Event board → missions and milestone track → relevant existing gameplay → consolidated claim → event shop/season track → next available activity.

From a contextual product: Upgrade need → Find Sources → self-chosen optional pack → store metadata and clear quantity → platform checkout → verified backend fulfillment → rewards delivered once → return to the exact suspended action.

14.2 Mandatory view states

Every screen requires a valid empty state, locked state with understandable requirement, loading state, offline state, ownership/completion state, pending purchase state, and recovery from cancellation/error. Small phones, tall phones and landscape/tablet layouts must remain usable. Correct navigation history matters: tapping Back from a chest should return to the inventory, not reset an unfinished board.

14.3 Screen economy matrix

Surface	Player's primary job	What the screen must never obscure	Legitimate commercial connection
Board	Arrange, forecast, activate	Board geometry, relevant objective, ports	Owned equipment applies only in correct mode; eligible boosters optional.
Results	Understand true outcome, claim reward	Actual resource flow, achieved target	Optional earned bonus or contextual next goal.
Empire map	See and expand owned city	Which business exists and is productive	Construction, buildings and capacity.
Business detail	Inspect and promote asset	Next benefit and exact requirements	Fragments, materials and source finder.
Manager roster	Assign, train and specialize	Actual manager role and skill utility	Recruitment, XP, books, equipment.
Recruitment	Spend earned ticket or selected currency	Probabilities, pity and price	Additional draws, documented milestones.
Equipment	Compare stat/ability changes	Material requirements and real benefit	Cores, parts and upgrades.
Research	Choose useful specialization	Branch prerequisites	Documents, time and speedups.
Contracts	Dispatch a valid blueprint	Duration, reward and slot eligibility	Slots, speedups, specialized materials.
Warehouse	Collect real production	Capacity, rate and elapsed time	Storage and efficiency.
Chests	Open or choose owned rewards	Full pool, saved item result	More earned/optional acquisition.
Merchant	Compare stock and reroll	Currency, limits, odds and reset	Voucher or Gem refreshes.

Surface	Player's primary job	What the screen must never obscure	Legitimate commercial connection
Collections	Assemble sets	True missing items and duplicate value	Source finder and choice boxes.
VIP	Understand level benefits	Permanent versus activation benefits	VIP XP and relevant packs.
Event	Participate and claim	Actual scoring rules, clock, earned route	Pass, event currency and eligible offers.
Shop	Browse optional goods	Real platform price, quantity and ownership	Commercial checkout.
Return report	Resume one useful goal	Previously earned progress	Genuine catch-up content, not forced upsell.

The technical annex lists the complete **41-screen implementation matrix** and individual state requirements.

15. Designing for commercial reality, not a theoretical "money printer"

The revenue goal is to maximize sustainable per-player contribution margin. The business will only work if acquisition costs, organic reach, retention, conversion and payer frequency permit it. A game containing all monetization mechanisms is **not evidence** those mechanisms improve outcomes. A tightly connected system is a hypothesis with many independent testable levers.

15.1 Revenue calculation

Expected contribution LTV = expected net IAP receipts + net advertising revenue − player-attributable backend, content, support, fraud and payment costs.

Then unit contribution after acquisition = expected contribution LTV − customer acquisition cost. Gross spending is not net earnings. Use cohort retention and payback windows rather than extrapolating a few early purchases into annual income.

15.2 Data to instrument from the first build

Capture first placement, first valid connection, first forecast improvement, first activation, first clear, time to understand multiple inputs, first business acquisition, first failed and successful construction, first manager assignment, chest open, first recruitment, merchant stock viewed, free versus paid refreshes, source sheet views, event progress, store details, checkout outcome, verified grant, first payer, repeat payer, refund, weekly return and long-term account milestones.

Never ship a randomized-paid feature with invisible probability logic or a store without funnel telemetry. Equally, do not instrument private personal data the game does not need. Apply appropriate consent, aggregation, deletion and regional requirements.

15.3 Important guardrail experiments

Test first-session clarity, progression rates, chest reveal speed, useful starter bundles, permanence of builder value, alternative free fragment rates, merchant stock diversity, actual pass completion and event overlap. Test retention, player trust and post-purchase churn alongside conversion. A short-term lift in ARPDAU can be harmful if it increases refunds or makes the free player population quit.

15.4 What Codex may implement, versus what remains unknown

Codex can build the engines, economy simulator, data catalogs, backend, IAP integration, content and visuals. It cannot establish a true best price or guarantee addiction, sales, user acquisition or retention without real human data. The system must be feature-flagged and observable so those questions can be answered by actual players and experiments.

16. Product acceptance: What "the final game" means

It is not enough for Codex to render a VIP screen or show a demo chest. A feature is done only if the entire action succeeds: eligibility → input → authoritative state transition → real reward or benefit → accurate UI feedback → persistence → correct reopening/retry behavior → accessible error and offline handling → analytics. Every important system requires at least one integrated journey and one failure test.

An example acceptance journey should take a fresh account from a genuine chain success to an owned Café, a queued construction, a completed dispatch, an earned chest, a fragment promotion, an activated event objective, a merchant refresh, a correctly displayed first-purchase offer, a completed purchase in platform sandbox, and a successful restoration on a second device. No fake payment or manually typed currency may stand in for this proof.

The original 104 challenge witnesses, economy invariants and legacy entitlements must survive the transition. Qualified paid-power boosts must never change the certified campaign result. A clean user with no purchases must be able to complete the 72 original core puzzles, obtain renewable resources and engage with representative manager, merchant, event and collection mechanics. A returning player must not lose purchased items after a network failure, reinstall or content update.

17. Editorial decisions that control the implementation

Firm design decisions: (a) preserve the existing skill-based chain board, (b) build persistent business ownership around it, (c) give rewards several meaningful acquisition routes, (d) share one rewards/items engine across shops, events and purchases, (e) introduce systems only after they become useful, (f) maintain a clear free progression path and a fair certified mode, (g) implement real event clocks and durable transaction authority, (h) make tactile feel a launch requirement, not polish afterward, and (i) design the catalog for extensibility rather than hand-authoring hundreds of disconnected packs.

Open tuning questions: exact prices, starter pack contents, franchise fragment curves, production yields, VIP thresholds, odds/pity balance, event point schedules, merchant stock and reset frequency, paid-power caps and which content generation should launch first. The technical annex supplies executable **starting hypotheses** and required simulations, not declarations of mathematically optimal figures.

Technical and release decisions: pure 2D Unity; Android-first; real Google Play IAP metadata and verification; server-backed valuable inventory; the original local puzzle remains playable offline; one canonical reward and ledger model; no public release, payments or ads activated until genuine store configuration, policy review and acceptance evidence exist.

PART II — NORMATIVE ENGINEERING, ECONOMY AND CONTENT SPECIFICATION

The following implementation annex is included **in full** so this document can be handed to Codex without relying on an earlier conversation. It contains the versioned item taxonomy, economy transaction invariants, starting loot tables, event/offer schemas, complete screen catalog, release checks, technical acceptance tests and implementation sequence. Part I controls the intended player experience; Part II governs operational correctness. If a numeric example conflicts with a later tested balance file, the versioned balance configuration controls the shipped value, and the design decision must be documented.

T00. CODEX EXECUTION CONTRACT — READ BEFORE EDITING

This is an **implementable product/engineering specification**, not permission to report success from mock screenshots, local-only billing or unchecked code generation. Codex must implement every subsystem, content definition, interface, event, migration, acceptance test and service boundary defined here. The work can be broken into sequential commits. Do not silently omit an item because it is expensive. Where a live external prerequisite is unavailable, finish the local implementation, tests, fixtures and deployment instructions, mark BLOCKED_EXTERNAL with the exact missing input, and continue independent work. Do not pretend the blocking prerequisite was satisfied.

1. Read AGENTS.md, STATE.md, PRODUCT.md, DESIGN.md, Docs/01_CASHFLOW_CHAINS_MASTER_SPEC.md, Docs/SHARED_TECH_MONETIZATION_LAUNCH.md, Docs/COMMERCIAL_IMPLEMENTATION.md, Docs/EXPANSION_AND_COMMERCIAL_PLAN.md and the **actual current sources**. Preserve original historical docs; write Docs/Decisions/2026-10-23-CC2-Commercial-Rebuild.md explicitly superseding incompatible rules (e.g. old ban on paid randomized rewards, server, ads, seasons) while **retaining** security, playability, accessibility and provenance requirements. Modify AGENTS.md to point at this specification. If source has changed from the audited baseline, reconcile differences and document them before editing.
2. Create git branch feature/cashflow-chains-2-economy; baseline commit, test report, baseline screenshots and reproducible current build. Never discard current authored puzzles, valid player saves, purchased entitlements or art rights documentation. Maintain a deterministic offline puzzle mode even when backend is down.
3. Develop in **gated vertical slices**, not a speculative one-shot rewrite. Each slice must compile, run, test, and generate actual screenshots. Ensure every advertised control operates; no fake products, invented prices, decorative buttons, simulated IAP success, invisible blockers or hardcoded countdowns in release.
4. Use strongly typed C# domain commands and events. Domain logic cannot depend on Unity UI. Presentation may never directly grant items, mutate purchased balances, decide random rewards, or alter authoritative outcomes.
5. Paid/economically valuable assets require server-authoritative state and transactions. Retain free offline puzzles; reconcile locally earned free progress via replay proofs with bounded, anti-replay backend acceptance. Never trust device clocks, client-selected premium rewards, client-calculated payments or client event scores.
6. All chance-based purchasable outcomes and paid random refreshes must disclose actual effective odds **prior to and close to purchase**. Clear real-money prices from platform metadata; unambiguous currency conversions and quantity. Paid premium currency does not expire. No redeemable cash, gambling payouts, fake urgency, misleading discounts, hidden automatic renewal, fake player activity or compelled ad viewing.
7. Build for adult/general audiences with an age-appropriate rating. Do not target minors with personalized high-pressure offers. Implement age/region policy feature flags and privacy/consent flows as applicable. All offers can be ignored without breaking ordinary puzzles.
8. Report scope, code changes, tests and failing tests, build IDs, captured screens, known defects, metrics, backend setup and external blockers after **every phase**. DONE requires all stated gates and platform dependencies, not merely successfully generating code.

T00.1 Specific baseline established from repository audit

- Domain sources: Assets/_Game/Scripts/Domain/Cashflow.cs (Pieces, Board, deterministic 8-beat Resolver, conservation) and Campaign.cs (CityRun, forecasts, missions, optional draft reroll).
- Existing: **32 piece recipes**, **72 free + 32 optional expansion challenges**, **16 flow-validated combo definitions**, **8 discovery sets**, **15 one-time route contracts**, **30 milestone achievements**; Unity 6000.3.26f1, URP 2D, Unity Purchasing 5.4.4 per repository snapshots.
- Key sources: Profile.cs, CommerceCatalog.cs (9 current product IDs), CityCollections.cs, Contracts.cs, Daily.cs, ExpansionCampaign.cs, MobileGameApp.cs, CityProgressionUI.cs, LocalStore.cs, PurchaseDelivery.cs, AtomicSave.cs.
- Current Profile schema 2 assumes small finite credit income, a limited receipt ledger, at most 104 level results, and narrow gem usage; it **must not** be extended by bypassing validation. Introduce an independent, versioned CC2 account/economy state and migration.
- Existing score from each puzzle is **not** autonomous idle income; persistent production is new and must be real, capped and separately accounted. Round cashflow is a **score**, not money to be withdrawn.
- Current IAP code is platform-sensitive and stores consumable state locally; use it only as an adapter starting point and reconcile server-authoritatively. Preserve ownership of cc.start.01, cc.blueprint.4, cc.skin.neon, cc.scenario.harbor, cc.scenario.finance plus any legitimately purchased consumables (where receipts and local history permit).

- Preserve Campaign.Get, ExpansionCampaign.Get and Resolver semantics for existing levels. No purchased power may silently alter certified original solutions, conservation or pre-existing results. The richer engine applies power in explicitly labeled empire/expedition modes, not fixed puzzle scoring.
- The screenshot's bright, lush, high-contrast miniature city, bold CTA and readable flow arrows are the **visual identity**, not proof of currently implemented economics.

T00.2 Definition of success and non-guarantees

Success requires: (i) fun tactile 2D puzzling, (ii) sustainable persistent progression, (iii) numerous meaningful free reward/paid acceleration paths, (iv) robust commerce and authoritative inventory, (v) evolving event content, (vi) measured player retention and monetization. No mechanic guarantees revenue. Optimize cohort contribution margin: LTV(IAP + ads) - acquisition spend - platform/payment fees - backend/support/LiveOps costs with retention and refund risk monitored.

T01. NON-NEGOTIABLE GAME VISION

Pitch: Build clever miniature-city economic chains, watch cash and goods pulse through them, then use their output to grow a permanent business empire: recruit executives, own/upskill franchise businesses, research technologies, operate deliveries, complete collections, explore rotating merchants and participate in live economic festivals.

Three coupled experiences:

1. **Moment-to-moment play (30 seconds–5 minutes):** drag, snap, rotate, reroute, forecast and activate 8-beat flows. Skill, experimentation, surprising but explainable combinations and lush tactile feedback remain the core entertainment.
2. **Persistent operating empire (minutes to months):** owned businesses, levels, stars, equipment, manager recruitment, tech, construction slots, dispatch contracts, storage and capped passive output. Upgrades influence eligible non-certified empire sessions/expeditions; original puzzle campaign stays fair and deterministic.
3. **Recurring LiveOps (days to seasons):** quests, collectible albums, weekly challenges, prize tracks, VIP, event shops, rotating merchant inventory, seasons, special generations, asynchronous leagues and cooperative projects. One valid action can advance several tracks.

Core principle: Every significant game action produces relevant gameplay/collection/progression/reward feedback. Every permanent goal has at least one real free acquisition route; premium products offer speed, extra capacity, flexibility, cosmetics or stronger eligible account development. The game must remain enjoyable if a player never purchases or watches an advertisement. Not every tap opens commerce; commerce is **available contextually**, not interruptive by default.

T01.1 Preserve versus change

PRESERVE: deterministic 8-beat solver and trace; conservation and caps; 72+32 puzzles and their legitimate ownership; real connection logic; free forecast and undo; building/art/audio assets and visual identity; city/contract concepts; responsive Android UX and accessibility; atomic local saves; test harness.

REBUILD: finite credits-only progression; one-time contract economy; flattened profile; hardcoded nine-SKU shop; narrow gem sinks; discovery-only collections; generic result screen; non-replenishing reward surface; hardcoded monolithic mobile UI controller; lack of server config, telemetry, content versioning, live offers and event engine.

T01.2 Modes (explicit paid-power boundaries)

Mode	Eligibility/power	Rewards and purpose
Certified Campaign	Fixed challenge board and owned-free rules; paid power excluded	Original 72 free + 32 owned extra levels, stars, first-clear rewards, mastery.
Open City / Empire Operations	Owned businesses, managers, equipment and researched perks apply	Repeatable cash/materials/XP within caps and daily budgets.
Dispatch Contracts	Send **validated saved layouts** into timed renewable commissions	Materials, tokens, fragments, cash, executive experience.
Daily Skill Puzzle	Same predetermined puzzle for everyone; paid power excluded	Earned reward, skill ranking where server verified, free entry.
Expeditions / Festivals	Optional limited repeatable objectives; owned development may count	Seasonal rewards, event-score contributions and tokens.
Cooperative Projects	Asynchronous verified resource/effort donations	Shared milestones and alliance gifts, no simulated members.

Mode	Eligibility/power	Rewards and purpose
Empire League	Explicitly progression-influenced leaderboard	Account power competitions; separate from equitable skill leaderboard.

No invented mobile real estate or tradeable actual financial product; all currency is purely fictional gameplay value.

T02. THE ECONOMIC NETWORK: SOURCES, ROUTES, SINKS, REWARDS

The player-centric graph is:

meaningful action -> deterministic result -> GameDomainEvent -> RewardTransaction -> inventory/progression/event/collection/pass/VIP -> new unlocked objective -> acquisition routes (play, wait, exchange, earn, random draw, optional pay) -> action.

Each item is a canonical ItemDefinition. Each acquisition route is a SourceDefinition. Each consumption is a SinkDefinition. Each reward container is a LootTableDefinition. Each real-money arrangement is an OfferDefinition. **Do not conflate these.** The *same* franchise fragment can be obtained through a contract, merchant, choice chest, random chest, event, alliance milestone and purchase.

T02.1 Mandatory currency/item taxonomy

Canonical ID / display	Free sources	Principal sinks	Rules
CASH / City Cash	Puzzle completion grants; capped open-city production; dispatch, quests	Buildings, research, upgrade fees, trades	Inflation-sensitive; cap and tune sinks. Distinct from ROUND_SCORE.
GEMS / Spark Gems	Achievements, events, pass free milestones, very modest login/quest earnings; IAP	Recruitment pulls, paid merchant refresh, flexible speedup, selected packs	Purchased Gems cannot expire. Show exact gem price; no client authority.
WOOD,STEEL,PLANS / construction materials	Production, dispatch, puzzle rewards, chests, merchant	Build/upgrade business/facilities, advanced construction	Three materials max in early game.
RESEARCH_DOCS	Contracts, free event missions, chests	Technology projects	Separate research time.
BIZ_FRAG:{id}	Franchise contracts, chests, collections, merchant, events	Business acquisition and star upgrades	Specific to business ID; duplicate conversion.
BIZ_FRAG_UNIVERSAL:{rarity}	Rare milestones, eligible events, choice chests	Eligible selected business fragments	Conversion rate explicit and limited.
MANAGER_FRAG:{id}	Recruitment, contracts, event awards, exchange	Acquire/promote manager	Prominent collection progress.
MANAGER_FRAG_UNIVERSAL:{rarity}	Free milestone, shop, premium tracks	Selected manager promotions	Universal substitutes scarce and valuable.
MANAGER_XP	Ordinary use, training contracts, chests	Level manager	Subject to level cap per star rank.
SKILL_BOOK	Events, dispatch, chest pools	Manager active/passive skill levels	Costs scale distinctly.
GEAR_PARTS,GEAR_CORES	Salvage, expedition, chests	Equipment improvement and mastery	Rare cores later game.
CONSTRUCTION_SPEEDUP_MIN	Earned missions, VIP gifts, event rewards, packs	Construction time reductions	Time denomination recorded in minutes.
RESEARCH_SPEEDUP_MIN,DISPATCH_SPEEDUP_MIN	Similar	Specialized timer reductions	Never secretly interchangeable unless GENERAL_SPEEDUP_MIN.
RECRUIT_TICKET	Onboarding milestone, missions, events, choice packs	Manager recruitment	1 ticket = 1 draw; no cash-out.
MERCHANT_VOUCHER	Daily missions, merchant	Random shop reroll	One initial free daily reroll by rule,

Canonical ID / display	Free sources	Principal sinks	Rules
	exchange, events		escalating paid extra refresh if active.
CONTRACT_TOKEN,ARENA_TOKEN,ALLIANCE_TOKEN	Respective activities	Respective exchange shops	Not broadly convertible to Gems.
EVENT_TOKEN:{season}	Event tasks and passes	Time-bounded event reward shop	Distinguish earned temporary tokens from any paid Gem balance. Explicit expiration on earned event-only tokens.
VIP_XP / Influence	Limited gameplay milestones and eligible purchases	Permanent VIP level advancement	This is XP, not a spendable coin; no decay.
COLLECTION_FRAGMENT:{album}:{piece}	Pack opening, expedition, event, gifting where allowed	Collection completion/exchange	Rare trade restrictions explicit.
THEME_TOKEN	Play/collections and cosmetics bundles	Non-power skins and personalizations	Cosmetics are genuine visible upgrades.

Avoid exposing all currency names at once. HUD default: Cash + Gems, current limited event currency if relevant; others live in context/inventory.

T02.2 Wallet/ledger invariants

- Every item grant/spend is LedgerEntry{entryId,accountId,itemId,delta,reason,sourceId,correlationId,configVersion,createdAt} with unique idempotency key. No negative spend balance, no unbounded integer overflow, no doubly claimed mission/receipt.
- ROUND_SCORE is a computed puzzle output; CASH only changes through explicit RewardPolicy after eligibility and replay verification. Do not grant unlimited cash proportional to arbitrarily repeated easy-score runs.
- Server stores purchased versus earned balances where laws/expiry/transfer rules differ. Purchased Gems always spent last or first under a published, consistent rule; no expiry. Never consume bought Gems for a free-only reward without explicit action.
- Award on complete/claim is atomic: ledger + inventory + milestone progress + relevant counters in one transaction. Event listener retries are idempotent.
- Economy events emitted once with stable roundId, contractId, chestOpenId, purchaseTransactionId or rewardClaimId.
- Offline proof: signed/challenge seed + board placements + base content version; replay Resolver server-side or in a trusted verifier; require budget, legal pieces, timeline and mission constraints. Rate-limit rewards and disallow editing stored result IDs.

T02.3 Acquisition route matrix — must be implemented

Every important scarce material requires 2+ non-purchase routes (at least one repeatable, sustainable route, except explicitly cosmetic/exclusive items). Source finder shows earned activity, shop currency, random chest odds, event availability and optional offers in that order. Source finder must filter impossible routes and show next availability accurately. No dead-end 'GET' button.

T02.4 Sinks and coupled value

- Faster construction increases material consumption, which increases reward demand; do not fabricate scarcity by secretly throttling after purchases.
- More contracts raise resource opportunities subject to dispatch capacity and daily server cap.
- More business stars unlock ability breakpoints and higher content; managers/equipment complement business quality rather than replicate it.
- Manager gear uses its own parts and cores; skill books separate skill progression from XP.
- Research improves defined systems through a transparent stacking and cap formula.
- New generations add new **optional** advanced progression and controlled new resources, not mandatory replacements that invalidate all existing heroes.
- Coins gained by repeatable play need renewable sinks: dispatch costs, construction, research, crafting, trader exchanges and maintenance of optional operations (not punitive loss of owned items).

T03. PERSISTENT EMPIRE — EXACT FEATURE SPEC

T03.1 Distinguish recipes, owned assets and board instances

- PieceDefinition = deterministic existing recipe: home, cafe, farm, warehouse, etc., with graph inputs/outputs and legal ports. - BusinessDefinition = progression metadata for a franchise type: rarity, role, unlock and fragments, upgrade tree, production outputs, optional active ability, cosmetics and permitted modes. - OwnedBusinessState = player's level, stars, XP, equipment, skin and one or more location installations. - BoardNode = an instance of a puzzle piece. Certified puzzles ignore owned upgrade bonuses. Open-City operations attach snapshot of eligible owned benefits *after checking stacking caps*, never rewrite base Piece recipes.

For each existing 32 recipe: map into residential, retail, industrial, logistics, energy, civic and financial groups. Introduce 8-12 fully designed initial persistent business assets; retain the other recipes available on fixed puzzles and unlock permanent empire progression gradually. New

T03. PERSISTENT EMPIRE — EXACT FEATURE SPEC

businesses must add functional differentiation: extra throughput, longer permitted paths, better energy efficiency, conversion flexibility, contract yield, synergy tags or storage behavior. Avoid 32 reskinned +5% stat sticks. Strong uncommon business ownership starts via earned progress after the player understands chains; do NOT hand out max-rarity hero or franchise on installation / invent a Day 1 grant.

T03.2 Business progression model

BusinessState{id,owned,level,starRank,fragmentCount,abilityLevel,equipmentSlots,skinId,assignedManagerId,installationIds}. Rarity: Common, Rare, Epic, Legendary. Star progression is separate from level and requires specific or eligible universal fragments. Stars can unlock ability levels and level caps. Upgrade result and sources shown before spending.

Default tuning hypotheses (all numbers are content-config and must be simulated):

| Milestone | New fragments | Cumulative including unlock | Meaning | |—|—:|—:|—| | Acquire | 10 | 10 | Usable permanent business | | Star 1 | 10 | 20 | Efficiency breakpoint | | Star 2 | 25 | 45 | Additional operating capability | | Star 3 | 55 | 100 | Distinct special ability | | Star 4 | 110 | 210 | Better cross-system synergy | | Star 5 | 220 | 430 | Late mastery |

Upgrades take Cash + materials + queue time; first few are near-instant to teach mechanics, later timers meaningful. Level cap by star, with server-validated acceleration. Rarity influences reward pools and resource requirements but must not create an inaccessible required certified puzzle.

T03.3 Persistent buildings and production

Persistent Empire map with 6–12 initially available plots (more unlockable). Install franchises and facilities. Each facility ConstructionProject{startedAt,finishAt,config,queueSlot}. Base 1 builder; earn temporary speedups; sell/earn a permanent second builder, later carefully gated further capacity. 1–2 freely unlockable plots early, additional via milestones. Construction can be started during simultaneous play; no mandatory puzzle wait.

Production model: each installation produces bounded material/cash output while active, subject to hourly rate, storage cap, production eligibility and **maximum 8 hours of offline accumulation by default**. Calculate in trusted service using timestamp and facility state; avoid infinite passive compounding and reject wall-clock tampering. UI shows real rate, capacity, accumulated output and when storage fills. Collect is a satisfying claim action, but support convenient batch-collection and optional auto-collection without using intentionally tedious micromanagement as the primary paywall. Persistent economy must never permit value overflow or exponential runaway.

T03.4 Managers (hero analogue)

Initial pool: 12 named original managers, 4 basic/common, 4 rare, 3 epic, 1 legendary; additional generations data-driven. Actual names, voice and illustrations original; no copied Whiteout/Kingshot characters. Roles: Operator (throughput), Developer (build), Financier (cash/research), Logistics Chief (dispatch), Merchant (exchange), Specialist (conditional new graph abilities). Each manager has levels, star ranks, specific + universal fragments, XP, two skill categories (operations/expedition), optional exclusive equipment at upper rarity, assignment limit and clearly described bonus formula. Starter manager earned through normal milestone, desirable higher-rarity manager acquired later through a guaranteed *free* progression reward or earnable recruitment, then developed through ongoing play. An introductory paid manager may be optional alongside the earnable manager; never falsely call it exclusive if available free.

Player assigns managers to owned businesses; disallow duplicate manager assignments where physically impossible. Unassigned managers retain collection/progression utility. Strategy: reassign based on contract type, business synergy or event requirements; presets save combinations. Guaranteed recruitment milestone provides reliable long-term target. Repeated manager level-up shouldn't require paid materials. Sell faster acquisition, equipment and flexible selection through multiple routes.

T03.5 Equipment and specialist materials

3 generic gear slots per manager at first unlock, expand to 4 only after engagement supports it. Gear rarities, levels, mastery and conditional set bonuses; stat caps and diminishing returns. GEAR_PARTS (frequent), GEAR_CORES (rare), duplicate salvage, crafted choice boxes. Equipment must visibly improve eligible output/dispatch/battle analogues, not certified puzzle score. A special tool for logistics may increase delivery slots; a finance tool may improve contract profitability. No gambling-dependent indispensable tool for basic play.

T03.6 Research and technology

Three initial trees: Operations (production throughput, storage, collection), Logistics (dispatch time, fleet capacity, contract returns) and Innovation (construction/research efficiency, higher progression access). Each node: costs, duration, prerequisites, explicit benefit tags, max level and cap. Show graph, currently affordable research and free versus paid routes. Research speedups and optional specialization. VIP/manager bonuses stack in *additive basis points with documented caps* (e.g. max 50% time reduction excluding explicit time-skip consumables) to avoid runaway synergy.

T03.7 Dispatch contracts and renewal

Replace 15 one-time commissions with recurring contract templates + rotating offers. Reuse ContractDesk's saved-layout verification and scheduled claim concepts. Contract eligibility requires an actually demonstrated chain/layout or sufficiently leveled business. Contract: id, template, tier, requirements, seed, cost, duration, yieldPool, assignedManager, dispatchSlots, eventTags, start, finish, claimId.

- Base 1 active dispatch, unlock second through free progression, optional purchased third or premium utility; cap according to economy simulation. - Free common contracts always available. Rare contracts can appear via rotating board or earned contract tickets; selective rerolls using vouchers/Gems only for optional better opportunities. - Durations vary by tier: initial 60s/3min/15min, then 1h/4h/8h for recurring

T03. PERSISTENT EMPIRE — EXACT FEATURE SPEC

progression; selected players may shorten via earned/purchased speedup, VIP or manager bonus. - One-click claim, reroute/restart, auto-restart convenience after legitimate unlock; no paid mandatory background feature for reasonable enjoyment. - Rewards determined on dispatch by versioned server rules or on claim by predetermined seed; never generated multiple times upon returning online.

T03.8 Storage and capacities

Expose exactly what limits production: builder slots, storage capacity, operating slots, active dispatch slots, management assignments, optional event attempts. Show both upgrade requirements and source finder. Limit scarcity to meaningful decisions, not arbitrary interruptions. Optional boost types: +production speed, +contract capacity, +research speed, +gathering/collection yield, temporary merchant discount, earned skill bonus. Keep certified skill mode neutral.

T04. UNIFIED REWARD ENGINE — CHESTS, GACHA, COLLECTIONS AND REVEALS

T04.1 Mandatory reward delivery modes

1. Direct deterministic item grant (cash/materials/fragments).
2. Earned randomized loot chest; 3–5 rarity tiers; table-driven weights; guaranteed rewards as separate line items.
3. Purchased randomized chest: exact odds disclosed nearby, explicit contents, server RNG and receipt-linked fulfillment; no hidden adaptive odds.
4. Rarity-themed containers (business, manager, equipment, research, event and collection chests).
5. Choice chest with allowed selection list and explicit quantities; consume chest only on confirmed selection.
6. Ten-open batch and fast-open with identical results/odds and full accessible result history.
7. Chest merge: choose to open N low-tier chests or exchange a disclosed quantity for one higher tier; verify expected-value fairness and exact distributions. No misleading visual odds.
8. Nested chest only if explicit nested probabilities and outcome disclosure supported; do not deploy if actual odds are too complex to communicate reliably.
9. Duplicate conversion (eligible duplicates -> universal shards or album tokens) with published ratios.
10. Fragment albums and collect-sets: partial progress, targeted earned exchange, optional gifting of eligible common fragments, rare trade restrictions made explicit; full set gives permanent modest bonus/cosmetic, not compulsory strength.
11. Claimable milestones (daily, weekly, account, seasonal, purchase-linked) separate from random draws.
12. Animation is a visualization of already committed reward result; reveal never determines outcome client-side.

T04.2 Initial chest definitions (simulator defaults; NOT real-money offers until approved, tested, odds displayed)

Bronze Supply Chest: Cash 50% (two quantity bands), WOOD 20%, STEEL 15%, PLANS 10%, common business fragments 5%; all outcomes values explicit. One guaranteed result. **Silver Operations Chest:** materials 45%, Cash 20%, Business fragments 20%, Manager XP 10%, Speedups 5%. **Gold Business Chest:** business fragments 50%, universal rare business fragments 15%, rare materials 25%, premium choice ticket 10%. Item contents and quantities data-driven; weight sum exactly 10,000 basis points. Rare reward highlight only on genuine high rarity. Tutorial grants earned bronze chest after first meaningful victory; premium equipment and advanced chest systems unlock later.

For a purchased randomized container, define exactly one disclosed item pool and quantity range per outcome; calculate effective odds for any multi-stage/nested draw. Pity protection (if any) must be real, stored, auditable and included in disclosures. Show item lists, duplicate rules, guarantee and any relevant regional availability before the purchase action.

T04.3 Manager recruitment / gacha (concrete default design)

Unlock only after completing initial core chain concepts and earning an introductory ticket (trigger-based, no fabricated calendar day). Recruitment RecruitBanner{id,activeFrom,activeUntil,cohort,lootTable,pityGroup,rateUpIds,featuredGuaranteeRules,earnedFreeDraws,eligibility}.

- Example banner odds per draw: 2.00% complete Legendary manager, 8.00% Epic manager fragments, 20.00% Rare manager fragments, 25.00% Universal manager fragments, 25.00% Manager XP, 20.00% Skill Books. **Total 100.00%; all exact quantities/featured sub-odds published**. This is a tuning starting point, not copied from another game.
- RECRUIT_TICKET or 150 Gems per draw; 10 draws for 1,350 Gems (illustrative). Earn at least one early ticket plus recurring free ticket sources; no perpetual obligation to buy to use recruitment.
- Every 10 draws: deterministic milestone chest; 30 draws: featured fragments; 60 draws: choice chest; guaranteed Legendary by 80 eligible draws. Pity counter carries within disclosed banner group; counter persists after logout/event transition according to explicit rule; no stealth reset. Guarantee upgrades *actual outcome distribution* and must be reflected on confirmation.
- Show visible 1/10/30/60/80 progress. Accept single/batch claims, duplicate recycling, draw log and replayable animation. Do not use paid gacha for an essential campaign solution.

- Draws can add progression to recruitment-related events, but avoid stacking so many mandatory queues that players cannot understand value.

T04.4 Free unlock with long development path

Teach basic business before first high-value earned unlock. Award an appealing Rare/Epic franchise or manager at a fixed **achievement trigger** after player has learned placement, rotation, building combination and a short contract. Give 10 unlock fragments/whole entitlement when milestone completed; show star progression and legitimate free sources, not a surprise paywall. Follow-on stars use fragment requirements in §T03.2 and optional targeted chest/merchant/pack sources. Maintain additional advanced layers: XP, skill, equipment, collection. This is a staged unlock and upgrade engine, **not** a Day-1 giveaway.

T04.5 Reward reveal UX

- Chests: squash -> lid unlock -> light expands -> weighted-color reveal -> item cards land -> inventory counters fly to destinations. 0.8–1.8 seconds default; Skip immediately reveals full true results; reduced-motion renders a calm static receipt.
- Single pull: anticipation limited and tasteful, rarity ring + sound chord, result history; no psychological deception like preselected near-miss symbols that imply false odds.
- Batch pull: tiles cascade in ~0.1–0.18s per item, expandable full details, one-tap Claim All, no dead waiting.
- Milestone: short, distinct claim animation; no synthetic delays solely to encourage further draws.
- Duplicate: original item visibly converts at published rate, not quietly discarded.
- Choice reward: side-by-side value and ownership context, clear confirm/back; no accidental spending.

T05. SHOPS, EXCHANGES, RANDOM STOCK AND CONVERSION PATHS

T05.1 One engine, multiple shop identities

ShopDefinition{id,availability,inventoryPool,slotCount,refreshRule,freeRefreshCount,refreshCostCurve,stockLimit,currencyTypes,eligibilityTags,expiryPolicy,purchaseLimits,oddsVersion}. Generate inventory **server-side** from auditable CSPRNG or signed seed; stock remains fixed until scheduled refresh or confirmed reroll. Paid rerolls always change shop offer sampling, but never guarantee better inventory unless explicitly stated. Show pool/rarity probabilities before paid reroll.

Create shops: Daily Merchant (Cash/Gems), Franchise Exchange (business tokens), Manager Guild (recruit currency), Equipment Workshop (salvage), VIP Merchant (prestige credits), Event Market (season-specific tokens), Alliance Exchange (group tokens), Collection Exchange (duplicates), Contract Board (variable offered missions). Only activate shops when their associated activities exist; they should not appear as ten useless unlocked buttons at once.

T05.2 Daily Merchant rules (default)

- 6 slots; seeded stock; daily UTC reset announced locally; no fake reset countdown.
- First *manual* refresh per reset free. Optional additional rerolls: 1 MERCHANT_VOUCHER, then 2, then 3, then cap 5 refreshes/day; if using Gems, explicit alternative cost 10/20/30/50 with cap. The player must choose source before confirmation, never silent fallback to premium Gems.
- Rarity pool displayed, no guaranteed upgrade except if defined (e.g. sixth refresh rare guarantee in a specific seasonal merchant). Favor deterministic first useful offers after unlock and varied subsequent stock.
- Purchase limits per item; notify at refresh not every action; stock purchase persisted independently. Lost internet means cached view read-only, not pretend transaction success.
- Offering purchased manager fragments in the merchant requires those managers actually usable; do not include future-generation junk.

T05.3 Exchange and sunk-cost balancing

The same item can be obtainable through different currency stores. No unrestricted arbitrage cycles (A -> B -> A with positive net balance). Build automated directed-cycle detection / bounded simulation. Define exchange fees/weekly caps where necessary. Paid Gems -> virtual items allowed; **no virtual currency -> cash-out**, crypto, external trade market or real-money redemption.

T05.4 Contextual Find Sources sheet

Any material icon is pressable and opens: amount owned, upgrade destination, amount missing, exact free routes, earned-currency stores, active reward pools with odds, next legitimate availability, and optional eligible direct purchase. Source ranking: relevant free playable action -> saved inventory -> earned shop -> earnable chest -> optional paid acquisition. If player has enough resources, primary CTA is Upgrade/Use rather than Buy.

T06. VIP, PERMANENT UTILITY, SUBSCRIPTIONS/CARDS, PASSES

T06.1 VIP / Influence 1–12

VIP permanent levels gained from gameplay milestones and qualifying purchases, with extra high-tier shop eligibility. VipTier{thresholdXp,benefits,freeDailyChest,eligibleOfferTags}; XP never expires. Benefits compound **within explicitly capped categories**; no arbitrary hard-coded percentage stacking. Initial design (test numbers, subject to economy sim):

Tier	Proposed new/stronger perk	Mechanically useful to
1	Claimable daily VIP gift	Collection loop
2	+2% construction speed	Build timing
3	+1 blueprint convenience preset	Layout management
4	+3% dispatch efficiency	Contracts
5	+3% production efficiency	Idle economy
6	+1 earned free dispatch slot via VIP OR same capacity from progression	Contracts; avoid exclusive lockout
7	Prestige merchant access	Dedicated currency spend
8	+1 manager preset	Assignment flexibility
9	Further +3% construction speed	Upgrades
10	Modest equipment-upgrade material discount	Equipment
11	Better daily gift, advanced trade choices	Late game
12	Prestige cosmetics and advanced shop inventory	Status and late progression

VIP activation voucher, if implemented, unlocks *extra* temporary benefits of achieved tier, not required access to already-earned core features. Free activation earned; Gems optional. Direct paid VIP XP in eligible bundles can be tested but exact total cost and free pace must be visible.

T06.2 Permanent utility

Prioritize Second Builder, extra Dispatch Slot, additional blueprint templates, batch collection/auto-claim, optional advanced manager preset, cosmetic city display slots. Each product meaningful even without other purchases. No paid undo, paid basic preview or paid correction of touch mistakes.

T06.3 Weekly/monthly card

- Weekly Operations Card: daily small resource bundle for 7 calendar claims; optional missed-day buffer/claim-all policy disclosed. Clearly **not** an auto-renewing subscription unless actually sold through Play subscription plumbing.
- Monthly Enterprise Card: Gems + daily material/utility benefit + automatic eligible routine claims or saved layouts, 30 days; no endless 'unlimited' claims. Calendar, daily benefit, expiration and platform billing model explicit.
- Paid premium Gems never expire. Time-limited card *benefits* may end per disclosed terms, but unspent purchased universal Gems remain.
- Nonrecurring card versus renewing subscription must be differentiated in product/checkout code. Support cancellation/account status, refunds, grace, and store-based subscriptions only if genuinely implemented and tested.

T06.4 Season pass

28-day initial season. 30 levels, rewards on **free and paid tracks**. XP earned from existing puzzles, contracts, business upgrades, manager training, collections, merchants and events; appropriate daily+weekly caps to avoid infinite farming. Premium purchase unlocks already completed milestones retroactively. Premium purchase never blocks free track. Support late entry catch-up, claim-all, expired-season handling and refund/ownership reconciliation. Offer tiers: basic pass and optional higher pass with clearly stated immediate tier skip, never compulsory to finish.

T07. LIVEOPS, EVENT STACKING, CONTENT GENERATIONS

T07.1 Multiple independent clocks

- **Account progression** (objective unlocks): chain tutorials, first contract, first manager, empire upgrades.
- **Account tenure** (truthful UTC elapsed): starter offer, comeback rewards.
- **Daily/weekly reset**: mission, shop refresh, VIP gifts, contract inventory; show accurate server time and local display.
- **Event cadence**: daily mini-objective, weekly theme, 28-day season, 4–8 week generation cadence *only after measured readiness*.
- **Content cohort age**: group of players receiving a shared progression season; do not assume an account is on 'Day 30' because a server is 30 days old.
- **Transaction eligibility**: first purchase, owned permanent unlock, pack limits, cumulative top-up milestone.
- **VIP/feature unlock**: thresholds and eligibility independent of install time.

T07.2 Event catalog: ship configurations and corresponding screens

1. **Construction Rush:** qualifying build starts/completions and used speedups; cap farming; economy materials, queue benefits.
2. **Franchise Festival:** acquire and star-promote actual owned businesses.
3. **Executive Recruitment:** draw, promote, improve manager skills; credit *eligible* account actions.
4. **Research Week:** completed research nodes; research documents and speedups.
5. **Delivery Derby:** valid dispatch arrivals and claims; route choice matters.
6. **Chain Masters:** score, actual resource throughput, specific earned combos; certified skill division no paid power.
7. **Merchant Festival:** verified exchanges and optional refresh mechanics; no requirement to purchase currency.
8. **Collection Expedition:** album fragments, complete themes, duplicate conversion.
9. **Equipment Workshop:** enhancement and set completion.
10. **Investment League:** account-development milestones and ranking with openly progression-influenced scoring.
11. **Shared City Project:** asynchronous real-player group projects, contribution caps and rewards.
12. **Seasonal Festival:** event tasks, chest keys, event currencies, featured generation rewards.
13. **New Manager Generation:** new earnable characters, featured banners, choice exchanges.
14. **Returner Catch-up:** reversible engagement aids after absence, no shame/punishment copy.

Each event definition contains eventId,contentVersion,cohortIds,openAt,closeAt,graceUntil,qualifiedEventTypes,pointFormula,milestoneRewards,freeObjectives,leaderboardDivision,shopId,passId,packTags,participationLimits,currencyExpiryPolicy.

T07.3 Event stacking rules

A single **verified** event (e.g. ManagerPromoted) can trigger permanent stats, daily objective, weekly event, season pass, collection, achievement and cooperative project, **each once**, using separate subscriber grants. Every subscriber must inspect eligibility and prevent replay. Never award points twice because an offer is purchased and the inventory is replayed; purchase milestone scoring and material-consumption scoring are distinct and transparently documented. Event countdowns genuine. Show relevant overlapping events without clogging reward UI. Include saving materials as strategy; no unexpected event retroactivity unless defined.

T07.4 Events as micro-economies

Support event keys, token shops, reward ladders, optional paid pass, event chest, guaranteed activity milestones, limited inventory, shared/individual rankings. Each sub-economy references the canonical inventory. Define currencies either persistent across reprises or explicitly earn-only expiring; no paid gems expiry. Reward loops cannot feed unlimited positive arbitrage.

T07.5 Generation releases

Server/config-cohort controlled introduction of new manager pool, 1–2 new businesses, research branch, gear tier and dedicated advanced crafting material only when sufficient player population reaches the prerequisites. Content is real and testable before event announcement. Existing veteran achievements remain relevant; catch-up path for new accounts. Feature flags for release, rollback and economy emergency pause.

T08. THE OFFER ENGINE — RELEVANCE, ROTATION, PURCHASE PATHS

T08.1 Data model and ownership

OfferDefinition{id,revision,platformSku?,offerFamily,bundleItemIds,quantities,choiceGroups,eligibilityRules,displaySurfaces,priority,validFrom,validTo,cohorts,perPlayerLimit,globalLimit?,cooldown,stepIndex,prerequisiteOfferId?,pricePresentation,promoEvidence,experimentId}. Platform SKUs configured in store; never invent localized currency strings. An offer may spend earned currency or paid Gems rather than require direct IAP.

T08. THE OFFER ENGINE — RELEVANCE, ROTATION, PURCHASE PATHS

OfferEligibilityService reads account unlocks, missing *useful* resources, relevant events, owned entitlements, stock, last presentation timestamps, locale, age/region restrictions and real store availability. OfferRanker scores contextual fit, not hidden vulnerability. A rules engine selects up to N contextual offers; no personal price discrimination; no changing paid loot odds by spend history. Keep all offer decisions audit-logged with reason and version.

T08.2 Catalog family inventory — mandatory engine support

| Family / ID prefix | Offer examples | Primary triggering context | Duration and alternatives | |—|—|—|—| | starter.* | $0.99 first-purchase combo of useful resources + original intro manager fragments | First meaningful empire progression; non-payer only | Permanent eligibility until purchased or declared event end; don't fake countdown | | permanent.builder | Extra permanent construction queue | Player has shown second viable project | Always available after feature unlock; earned temporary speedups alternative | | permanent.dispatch | Extra dispatch/ops slot | Regular concurrent operations | Permanent plus achievable free capacity over time | | permanent.blueprint | Extra templates / profiles | Existing free slots used | Existing purchased SKU restored | | gems.* | 4–6 top-up tiers | Gem balance view, eligible sources | Real localized tier SKUs | | daily.choice | Small choice chest / fragment selection | Targeted owned progression, reset | One/day stock, useful options | | daily.mixed | Hero, materials and equipment small pack | Progression + shop rotation | Daily; don't repeatedly interrupt | | weekly.card | 7-day operational resources | Renewable operations | Non-renewable card initial | | monthly.card | 30-day daily Gems/efficiency | Multi-week account use | Clearly shows terms | | growth.fund | Retroactive rewards at empire levels | Enough milestones completed | Purchase once; free base economy remains | | pass.standard | Season reward track | Season is active, player completing XP | Inherit completed milestones retroactively | | pass.premium | Plus specified level advance, not hidden subscription | Already understands season | Transparent alternative tier | | vip.milestone | Tier-special selection rewards | VIP threshold | Purchase limit per tier; permanent VIP unaffected | | biz.targeted | Relevant business fragments / choice pack | Owned/usable specific franchise under development | Free contract and merchant routes | | manager.targeted | Manager shards/XP/books | Eligible character exists or recruitment unlocked | Free recruitment / XP routes | | gear.* | Materials and choice gear | Gear available | Free salvage/events | | build.speed | Speedups + construction supplies | Eligible active construction | Free waiting remains usable | | research.speed | Research materials + speedups | Queued research | Free progression | | dispatch.speed | Shorten / increase dispatch opportunity | Active contract | Free wait and earned boost | | event.* | Tokens, event materials, optional attempts | Relevant event active | Explicit duration and limits | | event.staircase | Sequential $0.99/$4.99/$9.99/$19.99 tiers with free claim items between | Special declared event | Each next tier pre-disclosed, no misleading hidden price | | event.recharge | Guaranteed cumulative rewards for qualifying purchases | Promotional event | Transaction milestones logged, real calendar | | chest.choice | Known choice chest, not random | Multiple material bottlenecks | Guarantees exactly selected items | | chest.random | Purchasable loot container | Specific reward category, allowed region | Real odds displayed, pity if stated | | recruit.ticket | 1/10 recruitment draw currency | Recruitment unlocked | Earned tickets and free draw options | | merchant.refresh | Paid reroll voucher/Gem option | Current stock rejected, refreshes available | Disclosed odds / exact cost and cap | | collection.complete | Eligible fragment selection | Album nearly complete | Free trade/exchange sources | | cosmetic.* | Building art, particle trails, city themes, frames | Cosmetic editor/profile | Cosmetic only; no false stat claim | | prestige.* | Shop access/decoration choice | High VIP/late stage | Progression-based | | returner.* | Catch-up choice pack + useful earned missions | Actual prolonged account absence | No shaming or countdown without actual schedule | | advanced.* | Late-tier materials, specialized research bundles | Specific unlocked technology/generation | Never sell unusable materials | | expansion.* | Preserve existing Harbor and Finance puzzle packs | Challenge content progression | Existing entitlements honored |

Pricing test tiers (USD equivalent, **hypotheses only**, localized from Play metadata): entry $0.99; impulse $1.99/$2.99/$4.99; mid $9.99/$19.99; high $49.99/$99.99. App store product config controls actual price. Use reference value only if verifiable from exact non-discounted product within lawful promotion; do not manufacture "90% off" anchors.

T08.3 Dynamic pack lifecycle

Locked -> Eligible -> Available -> Displayed -> Opened -> Purchased/Declined -> Cooldown/LimitReached -> Restocked/Expired. A sequence-specific offer adds NextTierUnlocked transition. Lifecycle state persisted server-side. Product's contents displayed exactly and localized; current gear/generation substitutions must use versioned catalog snapshots per eligibility, not retroactively alter an offer already purchased. Track impressions and actions accurately (not every UI repaint).

T08.4 Rotation schedules and selection

- Baseline daily 4–6 merchant inventory slots refreshed at announced reset. - Weekly material/gear/manager pack families in a deterministic, server-configured rotation. Event packs exist only during genuine event windows. Group server/cohort schedule and individual eligibility both apply. - Level/unlock targeting: the same pack *template* can substitute eligible early resources with advanced resources only with explicit offerVariantId and exact contents. - New player pack after relevant feature: no Day 1 Legendary guarantee. Intro offer arrives after skill tutorial + first permanent objective. - Returning account: show concise Welcome back report with true elapsed capped production, catch-up missions and relevant products (at most one polite offer suggestion). - VIP threshold: unlock its exact special offer and shop inventory. Top-up milestone should be progression toward a published promotion; a purchase may earn normal contents + VIP XP + alliance gift + recharge progress **only where all four are genuine eligibility rules**.

T08.5 Contextual placement policy

| Surface | Context | Primary route | Optional premium route | Interrupt rule | |—|—|—|—|—| | Construction project | Missing STEEL or timer remaining | Contracts/production/wait | Materials/speedup offer | Embedded acquisition sheet; no modal after every build | | Franchise details | 25/40 fragments | Source finder / available chests | Targeted fragments / choice pack | Never auto popup | | Manager roster | Star cap / missing books | Missions/recruitment | Training pack | Contextual button | | Chest inventory | Earned chest count | Open chest | Obtain more eligible chests | No purchase prompt during reveal | | Gacha | Earned ticket shortage | Ways to earn ticket | Gem/ticket top-up | Odds displayed; user initiates | | Merchant | Unwanted stock | Wait/free daily reroll | Voucher/Gem refresh | Confirm price each paid reroll | | Contract list | All slots occupied | Wait, claim, free unlock | Optional extra slot/speedup | Relevant, no nagging | | Event | Near meaningful milestone | Play qualifying tasks | Event pack or pass | Event details shelf only | | Season pass | Free XP progress | Free track | Premium unlock retroactive rewards | On user-open; optional one-time preview after proven interest | | VIP | New tier reached | Claim earned gift | Tier pack |

T08. THE OFFER ENGINE — RELEVANCE, ROTATION, PURCHASE PATHS
Tier detail screen | | Result | Earned resources and next action | Claim, continue, choose | Optional rewarded ad for bounded bonus | One opt-in surface, not blocking Next | | General Store | User deliberately browses | Full catalog | Any eligible SKU | No interruption |
Offer presentation frequency budget: No mandatory purchase prompts during first 3 successful puzzles. At most 1 unsolicited offer introduction per session after relevant unlock and no more than 1 per 24h in early lifecycle, remotely tunable. Contextual static buttons do not count. Never cover active board manipulation, chain animation or earned reward claim with real-money offers. User can decline in one clear action; declines are respected for reasonable cooldowns.
T08.6 Recharge/purchase milestone design
Offer guaranteed additional milestone gifts for aggregate eligible purchases during a **truthful** event: e.g., purchases 1/3/5 qualifying transactions or clearly disclosed currency spend amount (jurisdiction/age restrictions may disable). Cap rewards; no misleading 'free gift' where a purchase required. Purchase-related reward track cannot create an infinite grant loop; idempotency keyed to transaction IDs. Transactions also can grant alliance gifts via server where co-op feature is enabled; never fake other members' contributions.

T09. RETENTION OPERATING SYSTEM — RETURNING FOR ACTUAL VALUE

T09.1 Nested loops

Loop	Cadence	Player desire	Economy and purchase relationship
Action	Each touch	Precise control and immediate feedback	Core value of game; never charge for touch correction
Round	30s–5m	Watch skillful chain reaction, beat objective	Cash, material, XP, missions, earned chest
Short return	Minutes/hours	Collect completed build or dispatch	Optional acceleration/capacity
Session	5–20m	Finish several upgrade targets, collect milestones	Contextual resource acquisition
Daily	Real daily reset	Fresh merchant, missions, claims, stage	Earned tokens, optional pass progression
Weekly	7d	Focused event / collection / league	Event pass/eligible packs
Seasonal	28d	Complete reward track/album, new goals	Premium track/monthly benefit
Long horizon	2–6 months	Perfect build, VIP, managers, franchise generations	New resource sinks, specializations

T09.2 Daily return without compulsion

Morning-style report on next actual app open: finished queue, capped offline output, a ready contract, daily objectives, changed shop stock and valid event. Daily reward free claim, no broken streak or loss of purchased property for missing days. Weekly chest based on cumulative activity rather than punitive perfect attendance. Push notifications **opt in** per category (construction finished, event started, contract ready, season end), with quiet hours, limits and region-safe permission UX; no fake urgency or notification spam. Email not required for play.

T09.3 Daily/weekly missions

3 daily missions after unlock, involving different ordinary gameplay systems, plus 1 optional stretch mission. 5–7 weekly goals and seasonal pass XP. No duplicate credit for rapidly repeating trivial no-cost actions; use source/event quotas. First daily reward easy, second varied, third meaningful, weekly jackpot guaranteed from real progress. Progress tracked server-side and replay safe.

T09.4 Collections, completion and identity

Business catalog, manager album, city cosmetics, gear sets, chain combo encyclopedia, franchise sets and themed fragment album. Reward partial milestones *and* set completion; show missing pieces and credible sources. Preserve existing eight sets, add deeper fragment-based albums separately. Public prestige identity: cosmetics, profile frames, city skins, event-earned badges; no virtual-value redemption.

T09.5 Social/competitive progression

MVP: asynchronous opt-in score boards on a deterministic, server-verified challenge, and progression-weighted Empire League distinctly labeled. Required later-phase deliverable: real alliances (join/leave, shared progress, member roles, gifts from eligible purchases, weekly cooperative projects) with reporting, privacy and protection against duplicate or fabricated members. Public chat is an optional disabled-by-default module; if enabled, build moderation, blocking and reporting before launch. Players must be able to thrive in solo mode. Avoid pushy alliance obligations, fake war emergencies or fabricated ranks. Competitive paid-power modes explicitly separated from verified equal-rules puzzle leaderboards.

T09.6 Personalization and onboarding

Teach based on **completed capability triggers**, not invented Day 1/7/30 assumptions. Player who races ahead sees advanced tutorial earlier; a slow player never gets an irrelevant banner for a feature not unlocked. Account tenure, feature unlock, event cohort and transaction eligibility are independent clocks. Returning players see real catch-up features with honest descriptions.

T10. FIRST EXPERIENCE AND PROGRESSION UNLOCK GRAPH

T10.1 First 90 seconds

App loads immediately into existing tutorial board; no registration, paywall or optional shop distraction. First meaningful drag target in 5–10 seconds after control; first successful chain within 45s for reasonable first-time testers (measure). Show input/output paths and what changed. First result reveals **earned** Cash, small material bundle and a basic chest, not a top-tier manager. A 2–3-step coach guides second rotation and third combined chain. Undo/forecast remain free. No ad or purchase interstitial here.

T10.2 Trigger-based unlock sequence (hard dependency, not elapsed-day promise)

Gate ID	Verified trigger	Newly visible feature	Player-facing reward/case	Commerce visibility
G00	First app open	Fixed Puzzle Board	Interactive connection	None
G01	First successful chain	Earned result chest/inventory	1 Bronze Supply Chest	No real-money CTA
G02	Complete rotation lesson	Materials and empire hub preview	Starter materials	No real-money CTA
G03	Finish three tutorial concepts	Permanent starter franchise + first build	Initial plot and fast builder project	Basic Earn Sources / Store accessible
G04	Complete first persistent business construction	Dispatch contract board and first repeatable contract	Small free contract	Eligible builder utility appears *after* need demonstrated
G05	First contract completed/claimed	Manager introduction and introductory recruit ticket	Guaranteed useful starter manager via earnable onboarding path	Optional first-purchase starter offer becomes eligible
G06	Manager assigned to a business	Franchise fragments and star screen	Progress sample from real mission	Relevant fragments acquisition routes
G07	First eligible 2-star business or 3 contracts	Merchant stock + one free refresh	One useful shop discount	Optional further rerolls
G08	Qualifying 6+ puzzle outcomes / varied activities	Daily Missions, Collection completion	Free milestone chest	Daily deals, weekly cards optionally visible
G09	Empire Level 5 and manager developed	Recruitment banner + pity progress	Earned recruitment ticket	Gem top-ups/tickets if initiated
G10	First participating weekly event	Event board, event reward track	Free event milestone	Optional event pack and pass
G11	Demonstrated simultaneous tasks	Research, extra dispatch/builder	First research completed	Targeted acceleration

Gate ID	Verified trigger	Newly visible feature	Player-facing reward/case	Commerce visibility
		upgrade routes		
G12	After sufficient actual play and completion	VIP progress, growth fund, season pass	Earned VIP XP and free track	Relevant eligible card/fund/pass
G13	Late progression thresholds	Gear, advanced materials, prestige shop	New useful economy systems	Advanced packs only when usable

All gate thresholds are editable in versioned config, and tests guarantee no circular dependency or impossible free route. Player should experience interesting collection/rewards before being asked to buy gacha. Skill tutorials never teach 'pay to succeed'.

T10.3 Long progression

Empire levels 1–50 initial content (finite authored upgrade tables, no infinite overflow). Content-generation unlocks after level/research/cohort eligibility. First 72 certified missions remain free. Existing 32 paid expansion missions remain owned under original entitlement rules; do not charge again. New repeatable expedition seeds must have solveable cases. First 12 authored empire operations tuned manually; generated repeatable templates verified by property-based solver before enabled.

T11. FULL SCREEN AND NAVIGATION SPECIFICATION

Navigation: Bottom tabs Board, Empire, Managers, Events, Shop. Global top bar only Cash, Gems, current timed event if active, Inbox/Notifications and user settings. Secondary routed features accessible within appropriate tabs or context. No six-currency clutter. Every screen: locked/empty/loading/ready/error/offline/owned/refunded/region-disabled states; back navigation; safe area; Android Back; accessibility; honest progress; consistent close buttons.

Screen ID	Surface	Key actions and components	Acquisition/monetization relationships
S00	Boot / resume	Version check, offline mode, background sync	No buy prompt
S01	Tutorial board	Drag house, rotate, connect, forecast, activate	Unlock real earned result
S02	Main board	Existing 2D city, available pieces, actual ports, score preview, Undo, Activate	Booster inventory for permitted modes only; no puzzle paywall
S03	Inspection drawer	Inputs, outputs, level effects, related business	Source finder / legitimate upgrade
S04	Chain animation	True 8-beat flow tokens, coins, escalating but skippable reward animations	None during animation
S05	Result	Score trace, earned items, contract/event/pass progress, earned chest, Next	Optional reward ad if available and disclosed
S06	Empire hub	Persistent district plots, construction, ready collections, upgrades	Construction slots, materials, VIP
S07	Business page	Level, rarity, star bar, manager, gear, related contracts	Fragments, speedups, choice pack
S08	Construction queue	Active builds, real timers, scheduling, free/premium speedups	Extra builder unlock, materials
S09	Warehouse	Output rates, caps, Collect All, material inventory	Storage and optional collection conveniences
S10	Research tree	Prerequisites, costs, durations, effect details	Research sources and acceleration

Screen ID	Surface	Key actions and components	Acquisition/monetization relationships
S11	Contract Board	Real rotated offers, refresh, duration, dispatch	Vouchers, Gems, capacity
S12	Contract progress	Timers, assignment, claim, logs	Speedup and dispatch capacity
S13	Manager roster	Filter, assign, compare, upgrade, manager collection	XP, books, fragments
S14	Manager profile	Skills, gear, stars, sources, presets	Upgrade resources
S15	Recruitment	Banner, rates, pity, ticket/Gem draw, guaranteed milestones	Ticket/Gem purchases via platform store
S16	Chest inventory	Category, chest merge, open x1/x10, choice chest	Earn more chests/eligible bundles
S17	Chest reveal	Accurate reward cards, skip/fast, duplicates, details	No forced upsell mid-reveal
S18	Daily Merchant	6-stock, free refresh, paid reroll, limits, reset time	Earned Cash/Gems, voucher, paid Gems
S19	Specialty shops	Franchise/manager/gear/VIP/ event/collection/alliance tabs	Earned token + eligible materials
S20	Inventory/source finder	Owned, required, sources ranked, exact odds where relevant	Crosslinks to gameplay and optional eligible offer
S21	Collections/album	Existing 8 sets + fragment albums, trade/exchange, complete rewards	Pack acquisition routes
S22	Combo encyclopedia	Existing 16 real-flow combos and progress	Focus on fun mastery; optional customization
S23	Daily objectives	3 + stretch, claim, reset time	Event tokens, free Gems, chests
S24	Events hub	Active calendar, upcoming genuine events, eligible reward tracks	Passes, event items
S25	Event detail	Score formula, milestones, rankings, tasks, inventory	Matching resources, pass
S26	Season pass	Free/paid tracks, level claims, remaining time, ownership	Premium pass, retroactive claims
S27	VIP / Influence	Tier progress, earned bonuses, daily claim, prestige shop	VIP-eligible bundles
S28	Achievements	Existing 30 achievements plus new economic achievements	Deterministic lifetime rewards
S29	Asynchronous League	Fair skill board vs progression-weighted Empire League, verified scores	Optional progression (never alter fair mode)
S30	Cooperative Project	True group progress, contribution limits, gifts	No fake teammates; alliance purchase gifts only when deployed
S31	Direct Store	Price metadata, product gallery, permanent vs consumable, restore	Real IAP purchase sheet
S32	Offer Detail	Exact item contents, currency/price, term, odds if random, eligibility	Confirm/cancel
S33	Purchase history/restore	Pending, confirmed, restored, refunded, support	No fake success
S34	Inbox / return report	Ready builds/contracts, earned claims, patch notes	One contextual offer at most
S35	Districts/challenges	Current 72 core + 32 owned expansion, replay stars	Existing expansion ownership

Screen ID	Surface	Key actions and components	Acquisition/monetization relationships
S36	Blueprints	Save/load layouts/presets; original purchased extra slots	Permanent convenience
S37	Cosmetic editor	Owned skins, preview, filters, city palette	Cosmetic store
S38	Settings/Help	Motion, contrast, controls, audio, privacy, odds explanation, account link, notifications	No hidden promotional traps
S39	Account / cloud recovery	Guest identity, optional link, restore, export/delete as law requires	Durable recovery, no forced login before core play
S40	LiveOps Inbox/Error	Graceful offline, disabled offer, stale version, network problems	Clear retry, no purchase charged twice

Screen-density rule: Board remains dominant. Don’t cram all systems onto one page. New menu unlocks include short coached demonstration and one free benefit. Notifications never cover essential board input. Modal opens and back navigation must be reliable on Android gestures.

T12. TACTILE DESIGN: EVERY TAP, DRAG, SWIPE, REVEAL AND UPGRADE

A commercial economy will fail if the underlying interaction feels cheap. Existing image identity: lush green miniature blocks, teal road network, bright mint economic pulses, cobalt/orange typography, vivid gold activation. Maintain recognizable coherent style; no ripped reference assets. Design 2D sprites, shadows and highlights consistent with actual board coordinate system (no nonfunctional rendered fantasy perspective). All text stays true native TextMeshPro/uGUI, not baked into art. Support 720×1600, 1080×1920, 1080×2340, 1170×2532, tablet and desktop companion.

T12.1 Touch response language (initial targets, tune by device playback)

| Interaction | Motion & feedback requirement | Sound/haptic | Actual logic | |—|—|—|—| | Tap button | Down-state scaling ~0.96 in <80ms, elastic return ~160ms | Soft tick optional | Action starts on valid release; no false state | | Long press | Progress ring, contextual inspector without obstructing finger | Gentle subtle haptic | No purchase action initiated by long press | | Pick up building | Lift/shadow, ghost placeholder, legal-slot preview, finger offset | Pickup chirp | Board unchanged until valid drop | | Drag | 60fps where feasible; smooth movement; glowing live connections | Gentle subtle cue when valid snapping changes | Forecast can be incremental but final must match solver | | Valid drop | Snap to center; tiny rebound; connection lines light up | Two-tone snap + haptic | One atomic board command; saved | | Invalid drop | Ghost recoils and returns; exact reason label | Light deny tone | No currency or lost piece | | Rotate | Pivot 90° around center; ports update and routes animate | Click haptic | Exactly one board rotation per action | | Undo | Reverse motion/connection glow | Reversing click | Free; undo only board placements not irreversible paid spending | | Forecast | Animated potential flow, colored throughput, bottleneck warnings | Soft scan cue | Derived from real Resolver output | | Activate Chain | Golden button depresses; power surge follows actual routes | Short impact then rising tonal rhythm | Authoritative result computed and persisted first | | Each beat | Resource token pulse travels actual port path; readable cash counters | Pitch increases with real legitimate combo | Token count and speed true to event log | | Real combo discovered | Distinct mini splash, symbol chain and collecting ribbon | Unique success chord | ComboBook true condition only | | Result reward | Cards and materials settle into inventory counters | Collecting flourish | Transaction already granted once | | Claim chest | Chest opens with tier-specific light, no false preview | Chest tier cue | RNG result committed before reveal | | Manager recruit | Reveal rarity ring and character entry pose | Rarity signature | Server-granted true reward | | Upgrade building | Construction mini-animation, particles, tangible visible altered façade | Construction chord | Queue/result state authoritative | | Star promotion | Fragment burst fills star, ability highlighted | Tier promotion flourish | Cost and ability state exact | | Merchant refresh | Shelves flip/slide, new products reveal | Shuffle | Inventory change signed/paid once | | Currency gain | Small pickup trails toward correct counter | Short coin pitch | Numbers reconcile with server ledger | | VIP level | Crest fills and announces specific perk | Prestige sound | Tier rule applies exactly | | Daily reward | Calendar/day stamp and item flyout | Cheerful collect | One real claim per reset | | Error/offline | Clear inline explanation, retry affordance | No punishing noise | No false balances or frozen progress |

All animation values hypotheses, not rigid production performance commitments. Render functionally meaningful flows and hierarchy first, then decorative particles within measured frame-budget. Reduce Motion removes unnecessary shake/flash/long animation while retaining information. Support mute, haptics off, colorblind shapes/patterns and large accessible touch targets.

T12.2 Cashflow chain spectacular moment (core visual product)

Use deterministic trace to animate each 8-beat cycle as a cohesive rising chain reaction: colored material tokens (people/goods/energy) flow from producers; converters pulse when truly fed; cash signs travel to top HUD; actual profit nodes get gold glows; legitimate completed combos light a short icon ribbon; at payout, a satisfying receipt enumerates both round objective and actual persistent reward earned. Keep a fast-skip button. Avoid excessive VFX obscuring readability or adding input lag. An excellent chain should look and sound noticeably better than a weak one while reflecting actual economics, not fabricated jackpot values.

T12.3 Visual polish specification

- Build sprite/UI component atlas: nine-slice cards, highlighted buttons, default/rare/epic/legendary rim materials, progress bars, stock shelves, chest glyphs, currency tokens, character portraits, six or more building activity animations, VIP seals, collection album frames, actual outline indicator of connected ports. - Micro-cinematics: chain activation, meaningful discovery, first manager acquisition, first franchise star, rare chest, new research tier, first island expansion. Aim 0.4–1.5 seconds for ordinary actions, skippable for repeated cases. - Persistent map visual evolution:

T12. TACTILE DESIGN: EVERY TAP, DRAG, SWIPE, REVEAL AND UPGRADE
buildings get lights/rooftop details/vehicles/street life as real upgrades complete; a tangible city transformation reinforces investment. - Dynamic ambience: bounded pedestrians, cars following correct lane direction, warm windows, neon signage, cargo movement, warehouse forklift icon movement; not distracting under HUD, disabled under reduced motion and performance profiles. - Portraits/managers: visually distinct original silhouettes and roles, readable at 64–128px. Optional small idle animations; avoid expensive skeletal animation requirement until 2D pipeline stable. - Audio: original varied 4–6 layers per sound category to avoid repetition; sound bus independent for UI, coin, effects and music; use platform haptics sparingly. - Reward rarity colors + icons + text; visual semantics accessible beyond color. - Pressable shapes have hitboxes ≥44dp-equivalent, top HUD always legible, game board receives substantial vertical area; one-thumb options for rotate/undo.
T12.4 Interaction delight performance gates
Never block board drag on networking or store fetching. Drag feedback p95 <50ms on midrange Android test device, ideally 16.7ms frames at 60fps; accept stable 30fps on lower end without dropped inputs. Long-run 15-minute thermal/battery test. Control latency/perceived responsiveness matters more than count of particle systems. Test success/failure/audio-off/reduced-motion/short-screen every screen.

T13. INITIAL ECONOMY NUMBERS — SIMULATOR BASELINE, NOT PROMISES

Codex must implement these as editable balance-v1.json values and run the economy simulator before shipping. Values are initial design hypotheses, chosen for coherent relative scale, not live game market facts.

T13.1 First-session target rewards

- Certified first puzzle: ROUND_SCORE based on actual resolver; first-clear CASH +80, WOOD +10, 1 earned Bronze Supply Chest; never repeatable for unlimited Cash.
- Second/third tutorial first-clear: CASH +100 each, materials +10 and +15; unlock first permanent starter franchise + construction project (small materials provided by tutorial so no deadlock).
- First contract: 60 seconds, requires actual verified unlocked chain; yields ~60 Cash + 5–10 material units; first claim grants starter manager ticket or predefined free starter manager.
- Default open-city operations: reward based on verified puzzle/combo skill with a **daily reward budget**; after cap, allow free practice and skill scores but no unlimited exploitable durable payout. Higher tiers expand budget through progress, not only payment.
- Starting empire: one builder, one contract slot, storage good for a few play sessions, a small manageable number of plots; faster early builds so initial learning doesn't stall.

T13.2 Baseline construction progression

Facility tier	Example Cash	Materials (WOOD/STEEL/PLANS)	Base build time	Gameplay effect
1	100	10 / 0 / 0	30 sec	Initial production unlocked
2	240	20 / 5 / 0	3 min	+8% output or slot
3	550	35 / 15 / 3	12 min	New visual detail, +10% output
4	1,200	60 / 30 / 8	45 min	Operational ability/breakpoint
5	2,800	100 / 65 / 20	2 hours	Specialized production

Beyond tier 5, use validated authored cost curves and caps; no simple unbounded exponential without simulation. Onboarding may grant time-limited construction vouchers honestly. First 3 projects ensure clear progression in first 10–15 minutes without forced wait.

T13.3 Baseline distribution targets (to be empirically tuned)

- 5-minute new session: one meaningful board win, one claim/upgrade and clear next objective.
- 10–20-minute returning session: several puzzles/dispatch decisions, 1–3 visible progress milestones, at least one free acquisition route relevant to current scarce material.
- 7-day ordinary nonpayer: multiple permanent upgrades, at least one recruited/upgraded manager, several chest openings, one event pass free-track experience; no mandatory paid gates.
- 28-day engaged nonpayer: visible franchise and manager development, recurring merchant and season value, achievable rare reward, never permanently blocked.
- Payer: permanent builder and cards save actual time; fragments/gear packs offer calculable advancement; no product instantly finishes every content system.

Do not turn these targets into fake achievements in production: tune amounts/rates until genuine gameplay produces them.

T13.4 Simulation cohorts

At minimum simulate 10,000 synthetic accounts/cohort over days 1–180 with: new casual 8 min/day, steady casual 15 min/day, frequent 45 min/day, optimizer 90 min/day; for each, nonpayer, entry payer, occasional payer, repeat payer, high spender; plus hiatus/returner and server-newcomer to advanced cohort. The spending cohorts must model voluntary demand rather than assume purchases happen by fiat: separate no-purchase counterfactual and utility/price-sensitivity Monte Carlo scenarios.

Produce distributions P10/P50/P90: time to build/manager stars; cash stockpile; shortage frequency; frequency of no-action states; per-day premium gems earned/spent; event accessibility; investment outcomes; content exhaustion; purchase-savings magnitude; variance from gacha; number of simultaneous eligible offers; amount lost from inventory caps; probabilities and effective acquisition cost. Detect arbitrage/infinite rewards, paywall dead ends, unaffordable upgrades and disproportionate gaps. Raise alerts and reject catalog if strong issues exist.

T13.5 Formula and caps

Use integer/fixed-point basis-point mechanics; explicit rounding. For stacking, effective = base * (10000 + min(sumBonusesBp, capBp)) / 10000, unless a researched multiplicative system has a documented alternative. Build speed floor and cap; no 100% zero-duration accidentally. Active speedups subtract bounded remaining duration and write new server timestamp. All probability distributions in integer weights summing to 10,000; deterministic seeded RNG in tests, CSPRNG in authoritative production reward outcome, with replay/audit record.

T14. DATA SCHEMAS, API CONTRACTS AND EVENT FLOW

T14.1 New canonical domain separation

```
Assets/_Game/Scripts/
  Domain/
    Cashflow/            # preserve solver, Board, Pieces, conservation
    Economy/             # items, wallets, transactions, caps, sources/sinks
    Empire/              # owned business, construction, production, storage
    Managers/            # recruit, stars, XP, skill, equipment
    Dispatch/            # recurring contracts/queue/eligibility
    Rewards/             # chests, loot tables, choice, merge, duplication, pity
    Commerce/            # offer rules, pack composition, stock, refresh
    LiveOps/             # events, passes, VIP, generations, collections
    Social/              # leagues, alliances, group progress
  Application/
    CommandBus/          # typed commands, validation, idempotency
    EventBus/            # typed events, replay-safe subscribers
    ClientSync/          # online/offline coordination
  Infrastructure/
    LocalState/          # atomic save, schema migration and safe cache
    CloudApi/            # auth/account, secure REST/websocket if needed
    Purchasing/          # Unity IAP 5.4.4 adapter, server verification flow
    Analytics/           # consent-controlled typed telemetry
    RemoteConfig/        # revisioned content/catalog with rollback
    Clock/               # trusted time, reset windows
  Presentation/
    Board/ Empire/ Managers/ Rewards/ Merchants/ LiveOps/
    Store/ Social/ Inventory/ Navigation/ Accessibility/
  Tests/
    EditMode/ PlayMode/ ContentValidation/ EconomySimulation/
```

Backend (separate deployable service and migrations): account/auth, economy/ledger, inventory, progression, contracts/timers, loot/rng, commerce/verification, shops, liveops, cohorts, leaderboards, notifications, telemetry/admin, config.

T14.2 Example ItemDefinition JSON


```
{
  "id": "BIZ_FRAG:cafe",
  "kind": "fragment",
  "displayKey": "item.cafe.fragment",
  "rarity": "rare",
  "stackable": true,
  "canExpire": false,
  "maxStack": 100000,
  "eligibleTargets": ["business:cafe"],
  "sources": ["contract:cafe", "loot:bronze", "shop:franchise", "event:franchise", "offer:biz.cafe"]
}
```


T14.3 Example offer JSON (an IAP bundle; NOT priced inside game data)


```
{
  "id": "offer.starter.v1",
  "revision": 1,
  "family": "starter",
  "platformProductId": "cc2.starter.01",
  "items": [
    {"itemId":"GEMS","quantity":90},
    {"itemId":"WOOD","quantity":30},
    {"itemId":"MANAGER_FRAG:starter_operator","quantity":10}
  ],
  "eligibility": {"all": ["first_contract_claimed", "zero_completed_purchases", "manager_system_unlocked"]},
  "surfaces": ["store", "manager_sources", "first_purchase_intro"],
  "maxPurchasesPerAccount": 1,
  "offerCalendarId": null,
  "experimentKey": "starter_01",
  "contentVersion": 1
}
```


T14.4 Example loot table JSON (does not mean silently making it paid)


```
{
  "id":"loot.recruit.standard.v1",
  "revision":1,
  "drawCurrency":"RECRUIT_TICKET",
  "totalWeight":10000,
  "outcomes":[
    {"id":"legendary_manager","weight":200,"quantity":1},
    {"id":"epic_manager_fragments","weight":800,"quantity":5},
    {"id":"rare_manager_fragments","weight":2000,"quantity":5},
    {"id":"universal_manager_fragments","weight":2500,"quantity":5},
    {"id":"manager_xp","weight":2500,"quantity":100},
    {"id":"skill_books","weight":2000,"quantity":2}
  ],
  "guarantees":{"legendaryByDraw":80,"milestoneDraws":[10,30,60]},
  "oddsDisclosureKey":"odds.recruit.standard.v1"
}
```


The example uses abstract reward category IDs; the production table must expand these to actual eligible manager IDs, published category-to-item probabilities and exact quantities. If pity alters distribution, present effective pity conditions and guarantee separately, and audit every roll.

T14.5 Typed commands / events

Commands: SolveRound, ClaimRoundRewards, StartConstruction, FinishConstruction, SpendSpeedup, CollectProduction, UpgradeBusiness, AssignManager, PromoteManager, UpgradeGear, ResearchTech, StartContract, ClaimContract, OpenChest, MergeChest, ChooseChestReward, Recruit, RefreshMerchant, BuyMerchantStock, ClaimDaily, ClaimEventMilestone, ClaimPassReward, ClaimVIPGift, RedeemFragment, SelectCosmetic, InitiateIapPurchase, ApplyVerifiedIap, Refund/RevokePurchase.

Events: RoundVerified, ChainComboAchieved, CashGranted, ConstructionCompleted, BusinessUpgraded, ManagerRecruited, ManagerPromoted, ContractClaimed, ChestOpened, ShopRefreshed, DailyMissionCompleted, EventPointsAwarded, PassLevelReached, VIPLevelReached, PurchaseVerified, EntitlementGranted, PurchaseReversed.

Critical sequence: a command validates state/cost/config/trusted time -> one database transaction deducts cost/grants rewards/logs domain event -> async projection/notification consumes event idempotently -> client receives authoritative state diff and reward receipt -> animations render receipt. No server response => no false paid reward claim. Retried requests use same idempotency key and receive original result.

T14.6 API shape (choose proven provider; do not hard-code proprietary SDK calls without checking)

POST /v1/auth/guest
POST /v1/auth/link
GET /v1/account/snapshot?sinceVersion=N
GET /v1/config/manifest
POST /v1/puzzle/verify-and-claim
POST /v1/economy/command
GET /v1/empire/status
GET /v1/events/active
GET /v1/shops/{id}
POST /v1/shops/{id}/refresh
POST /v1/loot/open
POST /v1/recruit/draw
POST /v1/iap/validate-and-fulfill
GET /v1/iap/entitlements
GET /v1/ledger?cursor=...
GET /v1/leaderboards/{mode}
POST /v1/support/report-transaction

All mutation routes authenticated via rotating session token, TLS, rate limited, request ids, replay prevention and version checks; never ship service secrets in Unity app. Server clock authoritative. JSON schemas versioned. Backoffice requires separate privileged credentials. Avoid overengineering real-time sockets until truly needed; polling + refresh suffices for asynchronous model.

T14.7 Local state and migration

New PlayerProfileV3/EconomyV1 separated from old schema 2. Migrate stars, best scores, free/paid level entitlements, blueprint ownership, skins and documented consumable balances. Store migration snapshot/backups, checksum, one-time migration ID, restoration rollback path and automated legacy fixtures. Do not zero legitimately earned old credits without an explicit mapping: convert old credits to new Cash at a defined, logged, capped conversion and preserve their equivalent spending utility. Do not invent purchased balances with insufficient receipt history. Old offline saves remain playable if cloud unavailable; server sync conflict UX offers safe import and support path, never silent overwrites. Normalize old cc.* SKUs to new entitlement IDs without repurchase. Upgrade Profile.Validate carefully rather than simply raising max counters. Game board retains saved current round across app updates.

T15. PURCHASE FULFILLMENT, PLATFORMS, REGIONAL SAFETY

T15.1 Unity IAP and authority

Current LocalStore.cs and PurchaseDelivery.cs are a starting point. Unity IAP **5.4.4** uses asynchronous StoreController purchase callbacks and explicit order fulfillment. For new CC2 commerce:

1. Fetch real Play product metadata and registered catalog IDs, render only actual available prices. Purchase button disabled on missing/invalid product or offline billing. 2. OnPurchasePending(PendingOrder): do **not** immediately grant client currency. Send receipt/transaction reference to

T15. PURCHASE FULFILLMENT, PLATFORMS, REGIONAL SAFETY
backend. Backend authenticates with platform verification API and checks package/product, account binding, acknowledged state, transaction uniqueness and purchase status. Store exact transaction ID securely or digest plus platform proof as appropriate. 3. In one database transaction, create the grant ledger record and exact entitlement/inventory changes once; respond signed receipt/fulfillment status. Client updates from authoritative state and **only then confirms purchase** through Unity IAP with correct PendingOrder. A network crash before confirmation is redelivered, and duplicate transaction handled idempotently. 4. Deferred/pending payment: do not grant. Failure/cancel: restore UI, no penalty. Outages: keep purchase pending, reconcile later. Reconnect: fetch purchases + server entitlements. Refund/revoke: policies implemented with audit history and purchased entitlements adjusted; never silently confiscate more than is appropriate for refunded item. 5. Permanent entitlements restore across devices/account link; consumable receipt history tied to durable backend wallet. Reconcile server state over local display; no cross-device 'consumables may be lost' as the final architecture. 6. Do not mark purchase acceptance as complete until **real Play sandbox** runs for success, pending, deferred, canceled, duplicate redelivery, offline interruption, account change, restore, refund and stock exhaustion.
Use current Unity IAP v5 documentation, not old IStoreListener/v4 examples. Reference: https://docs.unity.com/en-us/iap/purchases and https://docs.unity.com/en-us/iap/payment-providers/fulfill-purchases.
T15.2 Paid randomized rewards / shop refresh
Google Play: paid randomized virtual rewards must disclose odds beforehand and close to purchase; official https://support.google.com/googleplay/android-developer/answer/9858738?hl=en. Apple (future iOS): loot box odds must be disclosed before purchase; https://developer.apple.com/app-store/review/guidelines/. Implement odds drawers on real-money-priced randomized packs **and** on indirect purchasable routes where currency can be bought. All shop reroll outcome pools/probabilities disclosed when chance is purchased. Eligibility/featured sub-pools/caps and duplicate treatment part of versioned manifest. Region/age policy gating configurable; disallowed mechanisms fallback to deterministic choice rewards. Never misrepresent rarity visual presentation. No wager/cash-out economy.
T15.3 Ads and rewarded value
Rewarded ads opt in, disabled until actual production ad SDK integrated, privacy/consent data flow documented. Possible placements: post-result +X bounded materials, one extra daily merchant voucher, single low-tier chest, modest contract-speed credit. No forced interstitial in tutorial/board, no double-skip trick, no misleading reward. RewardedAdCompleted requires trusted SDK callback and idempotent daily cap. Ads reward should be balanced against IAP without paywalling basic content; remove ad placements where privacy/region/child policy requires. Advertising income recorded separately from IAP and tested for retention harm.
T15.4 Privacy, children, taxes and refunds
No forced email/account for opening game. Optional sign-in and transparent data handling for account recovery. Consent policy and GDPR/India DPDP/US state requirements reviewed for actual audience and SDKs. Data deletion/export, opt-out personalized offers where required, support and refunds. Avoid individualized discriminatory pricing. No exposure of player phone/email/precise location to other users. Publish actual privacy policy and Data Safety based on installed SDKs, not placeholder text. Age gate where necessary for random paid items/ads; safety mode deterministic rewards if needed. Backoffice must not expose purchase secrets. No gambling-like cash redemption.

T16. ANALYTICS, EXPERIMENTS, MONETIZATION OPTIMIZATION

T16.1 Typed event schema and provenance

Every telemetry event: eventId,anonymousPlayerId,sessionId,clientVersion,configVersion,eventName,clientTime,serverTime?,cohortId,experimentAssignments,screen,context,properties,consentFlags. Sanitize PII; no raw purchase receipts, secrets or private identifiers in logs. Offline telemetry queue bounded, encrypted/appropriate, sends only with policy permission and handles deletion/opt-out.

Required event taxonomy:

Domain	Events
Acquisition/onboarding	install_open, tutorial_step_view, tutorial_step_success, first_chain_success, tutorial_abandoned, first_empire_unlock
Board/input	board_enter, piece_drag_start, piece_drop_valid, piece_drop_invalid, rotate, undo, forecast_open, chain_activate, round_result, combo_discovered, result_next
Empire	business_unlock, construction_start, construction_finish, production_collect, inventory_full, upgrade_attempt, upgrade_complete, research_start, research_finish
Dispatch	contract_offer_view, contract_start, contract_complete, contract_claim, contract_refresh
Rewards	chest_obtained, chest_open, loot_rare, choice_chest_select,

Domain	Events
	chest_merge, duplicate_convert, collection_complete
Recruitment	banner_view, odds_view, draw_start, draw_result, pity_progress, guarantee_claim, manager_promote
LiveOps	event_view, event_task, event_milestone, pass_xp_earned, pass_reward_claim, merchant_stock_view, merchant_reroll, vip_level
Commerce	offer_eligible, offer_impression, offer_open, offer_dismiss, purchase_initiated, purchase_pending, purchase_verified, purchase_failed, purchase_refunded, sku_ownership_restore, currency_spent
Retention/support	session_start, session_end, return_report_open, push_opt_in, sync_error, billing_error, save_recovered, account_linked, support_ticket

Event properties include logical screen path, item IDs (not arbitrary full payloads), actual price tier/ISO currency from platform, display context, resource shortage amount, earned/purchased acquisition type, time since milestone, valid experiment exposure. Impressions fire only when reasonably visible, not on screen construction or each frame.

T16.2 KPI dashboard definitions

- D1/D3/D7/D14/D30 cohort retention by installation UTC cohort; distinguish active app session and notification open.
- Puzzle tutorial completion, first chain median/p90 time, valid-drop %, drag abort %, forecast usage, median round duration and repeat plays.
- DAU, WAU, MAU, sessions/user/day, sessions duration p50/p90, eligible feature usage, contracts/session, chests/day, merchant visits/refreshes, early churn reasons.
- Pay conversion by day and feature gate, paying DAU, first-purchase conversion, repeat-purchase ratio, offer-impression-to-purchase, IAP ARPDAU, total ARPDAU, ARPPU, whale concentration (risk monitor), refunds, chargebacks, subscription/card renewals (if active), paid Gem outstanding liability.
- Cohort net LTV at D7/D30/D90 and predicted payback by acquisition channel. Report total gross IAP, platform fees, ad payout, refund reserve, backend cost and UA spend to measure contribution, not only gross revenue.
- Free/paying gap, fair-mode participation, season pass completion, purchase-related churn, self-reported unfairness, regret signals and support incidence.

No claim of 'maximize money' is meaningful without verified analytics. Build dashboards and cohort exports before paid scaling.

T16.3 Experiment design and guardrails

Run randomized, remote-config-assigned experiments that vary **clear utility and value**: construction timer pacing, chest-versus-choice reward mix, merchant stock variety, relevant pack compositions, first-purchase pack value, pass track attractiveness, onboarding trigger placement, rewarded-ad availability, UI/tactile presentation. Avoid undisclosed changes to paid RNG odds or manipulative hidden price targeting. Pre-register metric, guardrails, minimal sample/decision rule and time horizon. Primary metric net D30/D90 contribution/LTV; secondary conversion; guardrails D1/D7 retention, refunds, free-player completion, frustration, crash-free session rate and support.

Failed experiments rollback by config revision. A/B assignment stable per user/account. Zero-user trials and automated bot play do not validate revenue. For extremely low traffic, focus qualitative playtests and iteration rather than misleading significance claims.

T16.4 LiveOps/admin control panel

Authenticated restricted internal dashboard: publish/draft/preview configs; activate events/cohorts; set shop rotations; validate no invalid odds, impossible source chains or missing assets; preview all player states; pause a broken paid offer; revoke misconfigured event; examine ledger and support refunds; inspect content simulation alerts; manage localization; manual *audited* compensatory grants with operator credentials and approval rules. No debug admin economy bypass in public Unity build.

T17. CONTENT DESIGN, ART BIBLE AND RELEASE ASSETS

T17.1 Initial content minimum

- Existing 32 building recipes and 104 campaigns maintained; author 12 empire-operation templates and 20 repeatable constrained variants with solver coverage.
- 10–12 owned business progression definitions from existing art and 8–12 additional functional/visual upgrade states as budget permits; 12 original manager characters with distinct silhouettes/skill kits; 12+ equipment pieces and 3 sets; 30–45 research nodes across three trees; 15+ contract templates with tier variants; 8 original sets plus 2 fragment albums; 3 starting chest tiers + 4 specialty chests; 2 recruitment banner templates; 6-slot Daily Merchant; 5 specialty exchange shops; initial VIP 1–12; one 28-day season config; full event catalog from §7, with at least 4 active/validated templates on launch and remaining ready behind feature flags for measured deployment.
- 20+ real offer templates, shop variants and 4–6 gem/entry tiers; no fake SKU/price displayed. Contextual acquisition screens across all upgradeable resources.
- Full onboarding/returning player content, UI empty/error/locked states, iconography, new sound/VFX suite, localization string IDs, accessibility and 3 display aspect-ratio acceptance screenshots per major feature.

T17.2 Content templates and localization

Every entity id,displayNameKey,descriptionKey,assetKey,unlockConditions,dependencies,stats,resourceCosts,rarity,visualTier,sourceRoutes,featureFlag,contentRevision. All localization strings in tables, not hard-coded C# screen logic; English default, architecture enables others. Right-to-left and long-string overflow tests at least on subset. Art assets with ownership/provenance, original authoring; no directly borrowed Century Games UI icons or characters. Maintain Docs/ASSET_PROVENANCE.md.

T17.3 Cosmetic collections

Noncompetitive city themes (neon, cherry blossom, nighttime freight, bright commerce, luxury finance), building trims, trails, frames, emotes where social enabled, avatars and sound themes. Guaranteed contents for direct purchases. If decorative items carry stats in optional advanced features, they cease to be purely cosmetic and must be labeled plainly; keep launch cosmetics stat-neutral to preserve clarity.

T18. QUALITY TESTS AND FAILURE INVARIANTS

T18.1 Domain correctness

- Existing every 104 certified puzzle solution replay passes untouched after migration; existing tests and deterministic property suites pass (the repository previously reported 59 domain tests, but current exact count must be verified at baseline).
- 100k randomized legal boards/resolver runs or optimized equivalent stress test; conservation, no fractional/missing resource, no unbounded cycles, no duplicated one-time awards.
- Every config has valid IDs, no missing art/localization, no circular unlock dependency, no unearnable prerequisite, rarity weights sum exact, no source/sink arbitrage cycle, no duplicate SKUs/currency definitions, no negative or overflow prices, no invalid timer durations and no impossible free mode.
- Business/manager upgrade costs, equipment, research, VIP, temporary benefits, queue slots, capped passive accrual all have deterministic unit and integration tests. No over-100% timer discount; no unbounded production accumulation.
- RoundCompleted can advance eligible 6 subscribers without duplicate grants; repeated event delivery idempotent. Cross-reset and season rollover correctly archive progress and avoid duplicate claims.

T18.2 Economy and commerce tests

- Source finder always points to at least one valid earnable source for required nonexclusive progress.
- Seeded chest distribution across 1 million simulated rolls statistically agrees with configured probabilities within tolerances; test guarantees exactly at pity threshold and after restore, batch equivalence, duplicate conversion and correct table revision.
- Merchant refresh: first free, paid cost ladder exact, stock persist, cap, expiry, concurrency, odds disclosure and transaction rollback.
- Buy choice pack then pick variant; guarantee exactly selected grant, never both. Paid item refund and entitlement reconciliation audit.
- IAP: price unavailable, cancel, pending/deferred, duplicate callbacks, process death between ledger commit and IAP confirmation, store restore, rejected receipt, product mismatch, account switch, consumable repeat, refund, partial network outage. Never charge twice nor grant twice; visible feedback factual.
- Negative Gem balance, manually edited local profile, changed wall clock, API replay, concurrent server calls, falsified event score, legacy invalid save, account merge and partial cloud sync fail safely.

T18.3 UX and device matrix

- Actual device/emulator Android captures for first launch, every unlocked system, all currency shortage sheets, rare and normal chest, merchant before/after refresh, paid-product metadata unavailable, event ended, offline resume, purchased SKU owned, save migration, pending IAP.

- 9:16, tall, short, tablet, landscape desktop companion; safe area, Android Back gesture, portrait rotation and resume mid-animation. Hitbox and overflow scan, right-to-left test strings, reduced motion, color high contrast and low frame-rate testing.
- At least 10 real first-time participants for core onboarding usability iteration and additional paid/nonpaid testers for longer-term behavior. No self-reported fun assertion based solely on automated test success.
- Performance: crash-free sessions, p95 frame time under target device budgets, no memory leak after 60 chest opens/100 rounds, stress 30 minute play, network reconnection, background transition, thermal impact.

T18.4 Security and incident drills

Simulate backend outage after paid confirmation, payment provider outage, DB failover, mid-event config mistake, mispriced real SKU, odds manifest mismatch, state corruption, chargeback surge, time change, duplicated RTDN/webhook event, excessive requests, image asset failure and telemetry failure. Feature flag kill switch must disable affected paid offers safely while leaving ordinary game playable; support refunds and accurate affected-user audit.

T19. CODEX IMPLEMENTATION ORDER — COMPLETE DELIVERY PROGRAM

Rule: Every phase is a code + data + UI + actual testing vertical slice. Codex should implement sequentially, committing and updating progress report after each. A future phase may be feature-flagged but not silently skipped. Do not promise full game done after one generated pass. Perform all work that is technically available; stop for external billing/credentials/legal approval only when unavoidable.

Phase	Required code and content deliverables	Exit gate
P00 Audit and freeze	Baseline branch, precise git SHA, original docs, UI captures, all tests, existing store ids, source map, technical debt	Existing game compiles; baseline screenshots/build checksum captured
P01 Decision/spec migration	Updated AGENTS.md, superseding decision docs, implementation tracker, architecture diagram, catalog JSON schemas, source/sink graph	No unrecorded design conflicts; all content schema validation tests
P02 Economy foundation	Ledger, wallet, inventory, item catalog, RewardTransaction, source finder interfaces, versioning, idempotent domain command handling, legacy-save migrator	Credit/entitlement migration fixtures; rollback; 0 duplicate grants
P03 Empire vertical slice	Owned businesses, initial franchises, buildings/plots, one construction queue, upgrades, storage, real capped production; responsive UI	Certified puzzles unchanged, first empire loop playable fully offline sandbox/server test
P04 Contracts and research	Renewable verified dispatch contracts, capacity, trusted timers, research tree, speedups, dispatch screens	Free path to all first research/dispatch goals; all timer/clock tests
P05 Managers/equipment	12 original managers, roles, initial unlock, star fragments, XP, abilities, 3 gear slots, UI and actual effects	Clear business/manager benefit without breaking fixed puzzles
P06 Rewards and collecting	Chests, choice and merge, deterministic sample seeded tests, RNG service, album fragments, duplicates, animations and claim history	No missing reward, item loss or phantom outcomes; odds tested
P07 Recruitment/gacha	Intro ticket, banners, odds UI, guaranteed thresholds, pity persistence, batch draw, duplicate conversion, source finder	Exact effective odds, draw history, guarantee and refund-safe records
P08 Shops/rotations	Merchant stock, one free reroll, paid vouchers/Gems rerolls, specialty shops, price/currency distinction, expiry, stock, contextual acquisition	Stable server-side stock, concurrency and migration safe; no free-stock exploit
P09 Offers/VIP/cards	New Offer engine, product template library, dynamic eligibility, storefront redesign, VIP 1–12, weekly/monthly cards, growth fund, sequential packs, recharge rewards	Every SKU eligibility truthful; source finder integrates paid/free routes
P10 LiveOps/pass	Typed event subscriptions, event scoring, 28-day season, free/paid tracks, 14 event definitions, timed event shops, content generations, remote config	Event stacking idempotent; end/start/offline/late join cases tested
P11 Backend/billing	Deployed/staged account/economy/auth/config services, API, production-like datastore, server purchase verification, Play IAP integration, restore/refund flows, audited admin operations	Sandbox matrix and security QA complete; BLOCKED_EXTERNAL if Play account keys missing
P12 Analytics/admin	Typed event telemetry, dashboards, cohort metrics, config A/B testing, balance simulator, backoffice tools, consent handling	Metrics reconciled with server grant ledger; config rollback works
P13 Social and competitive	Verified skill leaderboard, Empire League, genuine asynchronous group projects and alliances/gifts, moderation/reporting	No fake users, duped scores or abusive purchases; solo player not disadvantaged from core progression
P14 Polished product UX	All 41 screen states, art/audio/VFX, first-session speed, haptics, reduced motion, content localization, accessibility, screen-density tuning	Captured real Unity/Android screenshots; gesture and frame-time gates
P15 End-to-end content/balance	Authored tier costs, 10–12 franchise lines, research, manager/gear progression, 12+ repeatable operations, events, all offer variants, art/QA	10k–100k simulations and 1–180 day economy reports pass; playtester issues addressed
P16 Release candidate	Play Console product IDs, billing/license/config, privacy/Data Safety, store assets, Android AAB, external closed test where required, 3 device types, support docs	Real testing and policy audit passed; operator explicitly authorizes publishing
P17 Post-launch instrumentation	Gradual rollout, live economy monitoring, retention and monetization experiments, refund/support alerts, safe hotfix	No unresolved severe economy exploit; data-driven iteration, no invented results

Subphases may overlap development for efficiency but cannot pass gate on unverified dependent components. Prefer a **small complete end-to-end slice** from gameplay -> reward -> inventory -> earnable resource -> upgrade -> offer -> authoritative fulfillment before proliferating 30 screens. After vertical slice is solid, generalize config and fill the required catalog.

T19.1 Acceptance packages Codex must generate

For every Pxx: Docs/CC2/PHASE_Pxx_REPORT.md with git SHAs, exact files touched, all source/asset IDs, compile/test commands and numerical results, fresh Unity capture paths for before/after at 3 screen sizes, blocked external credentials, current feature flags, known issues severity P0/P1/P2, and next changes. Update Docs/CC2/TRACEABILITY.csv mapping every requirement of this spec to feature flag, code path, test ID, screenshot proof, and launch state. Missing traceability = not implemented. Maintain Docs/CC2/DECISIONS.md, Docs/CC2/ECONOMY_TABLES.md, Docs/CC2/CONTENT_MANIFEST.md, Docs/CC2/BALANCE_REPORT.md, Docs/CC2/COMPLIANCE_CHECKLIST.md, Docs/CC2/RELEASE_STATUS.md.

T19.2 Build/test commands

Use original Unity test/build scripts (Tools/test.sh, Tools/build.sh linux|android, Tools/playtest.py, Tools/campaign-playtest.py, Tools/android-playtest.py) only after confirming current support. Add Tools/cc2_content_validate.py, Tools/cc2_economy_sim.py, Tools/cc2_legacy_migrate_test.py, Tools/cc2_contract_test.py, Tools/cc2_liveops_replay.py, Tools/cc2_store_mock_test.py (mock only inside

T19. CODEX IMPLEMENTATION ORDER — COMPLETE DELIVERY PROGRAM

editor/integration tests, never a fake provider in production player build). Pin Unity editor and packages, cap workers to avoid OOM, and never mark failures as success due to missing UI driver input.

T20. FEATURE-TO-MONETIZATION NETWORK TRACEABILITY

This is the reference map of **mechanisms**, not just named packs. Every row is delivered by a canonical feature above or explicitly deferred by a truthful feature flag; no separate bespoke code for every row.

Mechanism family	Mechanisms incorporated	Owned implementation
Free-to-paid entry	free useful unlock, low-cost first purchase, partial fragment unlock, free premium currency, introductory trial boost, initial free chest/draw, visible long upgrade track, alternative earned route	Tutorial gates, Starter Offer, Business/Manager Progression, SourceFinder
Time/capacity	construction/dispatch/research wait, speedups, general speedups, parallel queues, storage, production cap, auto-claims, contract speed and capacity	Empire, Contract, Research, VIP, Utility
Quality progression	levels, stars, rarities, gear, gear mastery, manager skills, ability unlock, research prerequisites, collections, late materials and content generations	Business, Manager, Equipment, Research, Generations
Chance	weighted chest, themed chest, featured banner, duplicate conversion, batch draw, pity and guarantees, milestone ladder, nested table optional, chest merge, choice chests, rarity reveals	RewardEngine, Recruitment, Collection
Random shops	timed stock, randomized shelves, one free reroll, voucher/Gem paid reroll, refresh cost escalation, disclosed probability, rarity guarantee where promised, limited stock, VIP shops, event shops, barter merchants	ShopEngine, OfferEngine, Inventory
Multiple currencies	Cash, Gems, activity credits, fragments, research docs, speedups, event tokens, VIP XP, trade exchange	ItemCatalog, Wallet, SourceSinkGraph
Offer packaging	starter, entry pack, targeted progression pack, generic resource pack, gem top-up, permanent unlock, choice pack, weekly/monthly card, season pass, growth fund, event bundle, cumulative recharge, sequential step ladder, bonus claim tiers, visual collection, location-bound acquisitions	OfferEngine, Billing, Cards, EventPurchaseTrack
Rotation and relevance	account unlock triggers, elapsed tenure, content cohort age, generation, VIP, previous purchase, owned item, resource shortage, scheduled event, regional price/availability, returner eligibility	EligibilityRules, RemoteConfig, TrustClock
Event stacking	quests, progress bars, milestone claims, free/premium track, purchase grants, event material usage, ranking, guild contributions, multi-event scoring for same action	EventBus, Passes, LiveOps, Social
Retention clocks	immediate satisfying interactions, result progression, minute/hour timers, daily mission/merchant, weekly events, 28-day season, content generation, persistent account goals, configurable legitimate push	UX, Timer, Daily, EventScheduler
Social/status	fair skill competition, progression-power	League, Social, Cosmetics, VIP

Mechanism family	Mechanisms incorporated	Owned implementation
	league, alliance gifts for eligible purchases, shared projects, avatar frames, city themes, VIP prestige and collections	
Friction and fairness	free core actions, free basic preview/undo, nonpaying progress, deterministic guarantee alongside RNG, legitimate free sources, purchases as convenience/acceleration, explicit paid power on noncertified modes	CoreModePolicy, ContentValidation, SourceFinder
Reward theater	snap/drag/rotate, chain activation, true profit particles, opening chests, rare rarity VFX, collection completion, multi-reveal, unlock animation, currency flight, success sounds/haptic	InteractionFeel, GameFeel, RewardFX
Data optimization	offer relevance, segment/cohort config, funnel analytics, player economy, price metadata, experiments, refund retention checks, audience/privacy restrictions	Analytics, Backend, Config, Backoffice

T20.1 Resource destination / action-to-progression matrix

Action	Immediate result	Secondary progress	Tertiary commercial entry
Move/rotate piece	Real score forecast changes	Skill understanding and combos	Eligible utility boosters only in empire mode, never paid undo
Activate chain	Round score and actual earned reward	Daily mission, franchise XP, event, collection, season XP	Event packs/currency, growth fund relevance
Finish build	New productive capability	Construction event and empire level	Builder slot, speedups, materials
Claim dispatch	Cash/resources/chest/fragment	Contract event, manager XP, weekly goal	More dispatch slots, speedups
Unlock business	New permanent asset	Album and future star goal	Fragment choice boxes, targeted packs
Promote business	Stat/ability breakpoint	Event, VIP progression (if defined), collection	Scarce fragments, research, equipment
Recruit manager	New management option	Pity counter/milestones/event	More tickets, gear and skills
Open chest	Items, rarity reveal, duplicates	Album/promotion readiness, chest milestones	Earned source finder, optional related bundle
Reroll merchant	Changed eligible stock	Merchant task, new selection	Refresh voucher/Gems
Finish research	Efficiency and unlocks	Research event and new tiers	Documents and speedups
Collect production	Real cash/materials	Collection target/daily task	Storage/efficiency utilities
Complete collection	Set reward, badge, bonus	Higher-tier album target	Choice fragment boxes/cosmetics
Reach VIP tier	New rewards/benefits	Prestige progress	VIP special pack, eligible merchant
Claim pass mission	Free reward plus XP	Pass level and event goals	Premium track if chosen
Purchase eligible pack	Exact bought items	Eligible VIP XP / top-up track / alliance gift	New progression needs, not automatic upsell

T21. FULL DEFINITION OF DONE

A system is **DONE** only when its free loop, paid/earned acquisition routes, UI states, accessibility, save/backend authority, balancing data, telemetry, failure modes and real tests exist. A mock character list or UI panel alone does not satisfy this spec.

Overall game qualifies as 'feature-complete RC' when:

- Existing 104 challenges, deterministic solver and original entitlements are preserved and tested.
- Persistent business empire, managers, equipment, research, renewable contracts, production caps, fragment progression, source finder, chest/choice/merge/draw/pity, merchant rerolls, VIP, passes, cards, genuine offer rotation and content generations all work together on authenticated durable economy state.
- Offer catalog and odds are accurate; every active purchase product verified and restorable, genuine Play product metadata, no mock success and accurate refunds.
- Every major screen above works for unlocked/locked/empty/offline/pending/complete/error state; every tap/drag/rotation visually and mechanically responds, no decorative pseudo-controls.
- New events actually subscribe to verified domain actions; one event can advance multiple eligible reward tracks but never duplicate currency grants.
- Retention features and notifications are meaningful, optional and nonpunitive; free players have non-paywall progression and a gratifying reward cycle.
- Backend audits, RNG manifests, economy simulator, cohort telemetry, content publishing/rollback, security tests, Play policy and privacy review are complete.
- Actual Android physical-device and Play billing sandbox acceptance plus external tester feedback exist (not just a simulation), and release has explicit operator approval.

Launch only after complete implementation is validated. An initial release can gate expensive social and late-generation features behind flags but the code, config, tests and safe activation path for those promised features must be finished if this is to be handed off as an implementation of *everything*.

T21.1 Explicit operator-blocked dependencies

Codex may require access/credentials for Play Console app/products and license config, production hosting and secret management, analytics/ad network IDs, privacy policy publisher, consent SDK setup, signed AAB/upload and physical device/human tester verification. Never invent them. Mark BLOCKED_EXTERNAL and ask only for the precise credentials/approval after the independent coding is complete. No publication until user explicitly authorizes it.

T21.2 Work products to return

1. Git branch and commits, passing build/test log, screenshot/video gallery, Android test AAB/APK and backend infrastructure/deployment scripts.
2. Economy content tables, generated source/sink graph, item/catalog schema reference, event/offer catalog, odds disclosure manifests, backend/API documentation, admin operation manual and migration report.
3. Simulated 180-day cohort economy report with notes on nonpayer advancement and pricing hypotheses, telemetry dashboards and declared unknowns.
4. Full QA/security/payment matrix and compliance checklist; explicit remaining BLOCKED_EXTERNAL items, no false production-ready label.
5. One-page operator runbook describing how to rotate packs, launch an event, alter reward tables, roll back unsafe configuration, issue auditable compensation and review revenue/retention.

T22. HANDOFF INSTRUCTION TO CODEX — COPY VERBATIM

Implement **CASHFLOW CHAINS 2.0** from CASHFLOW_CHAINS_FINAL_GAME_BIBLE.md against https://github.com/zaynvirkk/Cashflow-Chains. This specification supersedes conflicting old product/commercial direction while preserving existing deterministic puzzle outcomes, all existing earned progress, paid entitlements, UX accessibility and validated gameplay. Read all repository docs/sources, run baseline tests and create a safe feature branch first. Implement all required specification sections in ordered P00–P17 vertical slices, with actual Unity/C# code, data catalogs, original art/UI/audio, a secure authoritative backend, real compliant Play IAP integration, full economy and reward systems, contextual offers, rotating merchants, gacha/chests/pity, VIP, passes, events, social/competitive features, telemetry and simulator. Do not leave stub handlers, fake paid prices, decorative interfaces or skipped requirements. After every phase compile, test, run genuine gameplay, capture actual screenshots and report full test evidence and blockers. Do not claim publication, player retention or revenue without genuine evidence. Where external secrets, app-store configuration or human QA prevent acceptance, implement everything independent, mark the exact BLOCKED_EXTERNAL gate, and seek operator action. Do not publish to a store or change existing paid ownership without explicit authorization.

T23. REFERENCES AND RESEARCH BOUNDARIES

Repository-specific grounding: https://github.com/zaynvirkk/Cashflow-Chains, current baseline sources and Docs/ listed above. Cross-game inspirations are high-level commercial architecture patterns observed in Kingshot and Whiteout Survival: staged hero acquisition, upgrade shards, complementary stat trees, VIP, timed construction, active events, dual reward tracks, rotating merchant/reroll, reward chance, server-age generations, distinct acquisition currencies, purchase-linked

social gifts. **This game bible does not assert possession of either company's private monetization code, account targeting formulas or proprietary balancing.** All concrete Cashflow figures are initial game-design hypotheses to test, not factual claims about those games or guaranteed commercial outcomes.

Official platform guidance (check current rules again immediately before release):

- Google Play payments and purchasable random item odds: https://support.google.com/googleplay/android-developer/answer/9858738?hl=en
- Apple App Review Guidelines (if iOS later): https://developer.apple.com/app-store/review/guidelines/
- Unity IAP v5 purchases and fulfillment: https://docs.unity.com/en-us/iap/purchases and https://docs.unity.com/en-us/iap/payment-providers/fulfill-purchases

End of master specification.